Remote Sensing Object Detection Based on Detail-Semantic Decoupling and Multiscale Coordinate-Guided Semantic Enhancement

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***Abstract*—Due to the severe scale variation of targets in remote sensing images, the dense distribution of objects, and the fact that many small targets occupy only a very limited number of pixels, existing detection methods are prone to losing shallow details and suffering from insufficient low-level semantic representation during repeated downsampling. To address this issue, this article proposes DSH-DETR, a remote sensing object detection method based on RT-DETRv2 that incorporates detail-semantic decoupling and multiscale position-aware semantic enhancement. First, in the encoder, shallow features are split into a detail path and a semantic path, so as to alleviate the representation conflict of shared shallow features between detail preservation and semantic modeling. Then, a multiscale position-aware semantic enhancement module (MPSEM) is designed to improve the representation capability of low-level semantic features in complex remote sensing scenes through directional-aware convolution, multiscale context modeling, adaptive semantic interactive fusion, and coordinate attention (CA) calibration. Finally, a high-resolution semantic reconstruction module (HRSM) is introduced to generate additional high-resolution detection features, which, together with the original multiscale semantic features, form a four-scale detection framework, thereby enhancing the model's ability to perceive small targets and targets in complex backgrounds. Experimental results show that the proposed method achieves better detection performance than the baseline RT-DETR on the VisDrone2019-DET, RSOD, USOD, and RS-STOD datasets. In particular, it performs better on small-object detection metrics and under high intersection over union (IoU) thresholds, demonstrating its effectiveness in complex remote sensing scenarios.**

***Index Terms*—Coordinate attention (CA), multiscale semantic enhancement, remote sensing object detection, RT-DETR, small object detection.**

Received 16 July 2026; accepted 19 August 2026. Date of publication 24 August 2026; date of current version 10 September 2026. This work was supported in part by the Major Science and Technology Innovation Project of the Ministry of Science and Technology of China under Grant 2022ZD0115802, in part by the Research Project of Xinjiang Space-Air-Ground Integrated Intelligent Computing Technology Laboratory under Grant 2025A05-2, and in part by the Tianshan Talent Training Project Xinjiang Science and Technology Innovation Team Program under Grant 2023TSYCTD0012. *(Corresponding author: Zhenhong Jia.)*

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Digital Object Identifier 10.1109/TGRS.2026.3726514

I. Introduction

With the rapid development of remote sensing imaging technology, UAV platforms, and Earth observation systems, high-resolution remote sensing images have been widely applied in marine surveillance, traffic inspection, disaster assessment, military reconnaissance, and land-resource investigation. As a key foundational task in intelligent interpretation of remote sensing images, the performance of object detection directly affects the accuracy of subsequent scene understanding, target monitoring, and decision analysis. However, compared with object detection in natural scenes, remote sensing images usually feature dramatic scale variations, dense target distributions, complex background textures, and diverse target orientations. In particular, a large number of targets occupy only a very small number of pixels, leading to weak feature responses, indistinct edge textures, and a greater tendency for detail loss and semantic suppression during repeated downsampling, which poses greater challenges to detection models.

In recent years, deep-learning-based object detection methods have made remarkable progress in the remote sensing field. Convolutional neural network detectors represented by Faster R-CNN [1] and the YOLO series [2], [3], [4] have demonstrated good practicality in remote sensing object detection by relying on mature feature extraction capabilities and multiscale modeling strategies. To address small and dense targets, researchers usually enhance a model's representation ability in complex scenes through feature pyramids [5], [6], [7], receptive field expansion [8], [9], [10], and attention mechanisms [11], [12], [13]. However, the local receptive-field nature of convolution still imposes limitations on modeling long-range dependencies and global context, and this issue becomes even more pronounced in remote sensing scenes with large target-scale spans and strong background interference.

The introduction of Transformer [14] has provided a new research paradigm for object detection. DETR formulates object detection as a set prediction problem, uses global

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self-attention to model long-range dependencies, and removes complex postprocessing steps such as anchor design and nonmaximum suppression. On this basis, Deformable DETR [15] significantly improves multiscale feature modeling efficiency and training convergence speed through deformable attention. Furthermore, RT-DETR [16] is an important innovation in object detection. It discards the NMS postprocessing step used in traditional detectors and achieves both high accuracy and high speed in real-time applications by optimizing the Transformer architecture. RT-DETR not only outperforms YOLO-like models in computational efficiency, but also exhibits certain accuracy advantages, especially in complex scenes. Nevertheless, existing RT-DETR series models are mainly designed for natural-image scenarios. In remote sensing images, especially for small-object detection, they still suffer from insufficient use of shallow high-resolution features, limited low-level semantic representation, and inadequate collaboration between detail information and semantic information.

To address the problem that shallow information is easily lost in remote sensing small-object detection, existing studies have attempted to improve model performance by introducing high-resolution auxiliary branches, multiscale feature compensation, and attention-enhancement mechanisms. Although these methods improve the detection of dense and small targets to some extent, they still have two shortcomings.

First, Shallow features often simultaneously serve two functions: preserving local details and conveying high-level semantics. Existing studies have shown that object classification and object localization are not fully consistent in terms of the feature attributes they rely on: the former depends more on semantic discriminative information and contextual relationships, while the latter relies more on local cues such as edges, textures, and spatial structures [41], [42]. Therefore, if shallow features are directly shared by subsequent functional paths without decoupling, representation competition and optimization interference are likely to occur, which can weaken boundary responses for small objects and increase the risk of false detections in complex backgrounds. In remote sensing images, where targets are smaller in scale, background textures are more complex, and object distributions are denser, this conflict caused by shared shallow representations is often more pronounced, thereby limiting the expressive potential of shallow high-resolution features.

Second, existing semantic enhancement methods mainly focus on context modeling at a single scale or in a single form, and pay insufficient attention to the prominent directional structures, severe scale variations, and complex background interference in remote sensing scenes. As a result, they fail to fully exploit the advantages of high-resolution features for small-object detection.

To address the above issues, this article proposes DSH-DETR, a remote sensing object detection method based on detail-semantic decoupling and multiscale coordinate-guided semantic enhancement. Built upon RT-DETR, the method first constructs a detail-semantic decoupled dual-branch diversion mechanism in the encoder, where the same shallow feature, C3, is fed into a detail branch and a semantic branch, respectively. Through two parallel learning paths, D2 and S2 are generated, thereby achieving functional decoupling of the shallow mixed features and alleviating the functional conflict caused by shared features. Subsequently, a multiscale position-aware semantic enhancement module (MPSEM) is designed in the semantic path. By incorporating strip convolution, dilated convolution, adaptive semantic interactive fusion, and coordinate attention (CA) calibration, this module strengthens the joint modeling capability of low-level features for directional structures, long-range dependencies, and spatial positional information. Finally, combined with a high-resolution semantic reconstruction module (HRSM), the method generates additional high-resolution detection features and, together with the original multiscale semantic features, builds a four-scale detection framework, thereby improving the model's detection performance on small objects and targets in complex remote sensing scenes.

Experimental results on the VisDrone2019-DET, RSOD, and USOD datasets show that the proposed method outperforms comparison models on metrics such as average precision (AP), AP_{50}, and AP_{75}, with especially more evident improvements on small-object detection metrics, demonstrating its effectiveness in complex remote sensing scenes.

Compared with existing methods, the contributions of this article are as follows.

1) In the semantic branch, an MPSEM is constructed. Unlike existing methods that mainly rely on single-scale convolution, channel attention, or spatial attention for feature enhancement, this module integrates strip convolution, dilated convolution, adaptive semantic interactive fusion, and coordinate-attention calibration to jointly model directional structures, scale variations, long-range dependencies, and spatial positional information. As a result, it is better suited to the complex feature distributions of remote sensing images, which are typically characterized by large scale variations, dense object arrangements, and strong background interference.
2) At the feature output stage, an HRSM is incorporated. Different from existing multiscale detection frameworks that rely solely on the original pyramid features for prediction, high-resolution semantic representations are further reconstructed while preserving the original semantic hierarchy. These reconstructed features, together with the original multiscale features, form a four-scale detection architecture, thereby improving the perception of small targets, weak targets, and targets embedded in complex backgrounds.

II. Related Work

A. Remote Sensing Object Detection Methods

Remote sensing object detection has long focused on issues such as multiscale target modeling, dense target separation, and complex-background suppression. Among traditional methods, two-stage detectors represented by Faster R-CNN usually achieve high localization accuracy and strong feature representation ability, but their inference speed is relatively

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slow due to candidate-region generation and multistage processing. One-stage detectors represented by the YOLO series obtain higher detection efficiency through end-to-end direct regression and thus have clear advantages in engineering deployment. However, in remote sensing imagery, large numbers of small, densely arranged targets are easily disturbed by background textures, making one-stage methods more prone to missed detections, duplicate detections, and category confusion.

To address the characteristics of small targets in remote sensing images, existing studies usually improve detection performance from three aspects. First, shallow high-resolution feature representation is enhanced to preserve edges, textures, and local geometric cues, thereby improving the discriminability of weak targets. Chen et al. [17] proposed Freq-DETR, a frequency-aware Transformer framework for real-time small object detection in UAV aerial images. Specifically, a frequency-enhanced convolution module (FECM) was designed to extract spatial- and frequency-domain features while preserving edge and texture details through DTCWT-based high- and low-frequency decomposition. In addition, a decoupled spectral feature interaction module (DSC-Clo Block) was introduced to model high-frequency local information and low-frequency global information at higher level feature layers, thereby enhancing foreground-background discrimination with fewer parameters. Furthermore, an attention-guided selective feature pyramid network (AGS-FPN) was constructed to achieve efficient multiscale feature fusion by using high-level contextual information to select low-level features. It should be noted that the "decoupling" in Freq-DETR is essentially a high-frequency/low-frequency decomposition and interaction in the frequency domain. Its core objective is to preserve high-frequency detail information while modeling low-frequency global information, thereby enhancing feature representation and improving foreground-background discrimination. Such methods focus on how frequency components are organized and utilized, rather than on the optimization conflict of shared shallow features between the two functional roles of detail preservation and semantic transmission. Therefore, although Freq-DETR can alleviate detail loss caused by downsampling to a certain extent, it does not explicitly construct a detail path and a semantic path, nor does it specifically model the problem of insufficient semantic representation in low-level features. In essence, the "decoupling" in Freq-DETR refers to frequency-component decoupling, whereas the "decoupling" in this article refers to functional decoupling of shared shallow features. The two address problems at different levels. In summary, although existing frequency-domain decomposition or task-decoupling methods have certain advantages in detail preservation, context modeling, or head collaboration, they still do not explicitly decouple the functional conflict of shared shallow features at the encoding stage. As a result, they cannot directly solve the problem addressed in this article, namely, the conflict between detail preservation and semantic modeling.

Second, models are made more adaptable to scenes with severe scale variation by constructing multiscale feature fusion structures or introducing context-compensation branches. Gao et al. [18] proposed MSNet, a multiscale network for object detection in remote sensing images. First, P^2CEM is designed to extract spatial and channel features in parallel through partial convolution and pointwise convolution, thereby suppressing background noise while reducing computational parameters. Next, LGIFM is designed to capture texture information and semantic information through local-information stacking and context modeling, respectively, and to enhance target feature representation in the presence of small targets and complex backgrounds through weight allocation. Finally, LGIFP is proposed to combine CARAFE upsampling with LGIFM, enabling efficient cross-scale feature fusion and dense connections. Li et al. [19] proposed TBNet, a texture- and boundary-aware network for detecting small weak targets in remote sensing images. First, a texture-aware enhancement module (TAEM) is designed to explore and enhance texture details inside targets by capturing pixel-wise correlations, thus providing rich semantic References for classification tasks. Second, a boundary-aware fusion module (BAFM) is proposed to generate boundary maps using the Sobel operator and extract boundary features through asymmetric convolution, thereby strengthening spatial position information, especially for small weak targets with blurry boundaries. Finally, a task-decoupled RCNN (TD-RCNN) is designed to completely decouple the classification and regression branches: a decoupled sampling strategy (DSS) generates proposal boxes of different densities for different tasks, a decoupled proposal expansion strategy (DPES) introduces more contextual information for the classification branch, and a proposal synthesis strategy (PSS) generates high-quality samples for the regression branch. Unlike Freq-DETR, TBNet places greater emphasis on improving the discriminability of small and weak targets through texture and boundary information. Specifically, TAEM and BAFM enhance internal texture representation and boundary localization capability, respectively, while the "decoupling" in TD-RCNN mainly refers to separating the classification and regression branches, which is a form of task-decoupling optimization at the detection-head level. Although this strategy helps alleviate learning interference between different detection tasks, it is not equivalent to the functional decoupling of shallow features emphasized in this article. In other words, TBNet mainly improves texture discrimination, boundary localization, and the separation between classification and regression tasks, but it does not explicitly mitigate the representational conflict between detail preservation and semantic modeling from the perspective of functional division of shared shallow features, nor does it construct a dedicated enhancement path for weakened low-level semantics. Therefore, the "decoupling" in TBNet belongs to optimization at the detection-task level, whereas this article focuses on the functional branching of shallow features during the encoding stage between detail preservation and semantic modeling.

Third, false detections are alleviated by suppressing responses from complex backgrounds and enhancing the saliency of target regions. Hu et al. [20] proposed FSENet, a feature suppression and enhancement network for tiny object detection. First, a feature suppression and enhancement module (FSEM) is designed. In it, the feature enhancement module (FEM) strengthens deep feature representation by

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fusing shallow large-target features, while the feature suppression module (FSM) adopts a plug-and-play design and suppresses background interference and redundant information by subtracting deep large-target features from shallow feature maps, thereby alleviating the feature-mixing problem caused by feature fusion. Second, an improved Kullback-Leibler divergence loss function (IKLD) is proposed, which models bounding boxes as 2-D Gaussian distributions, normalizes them with the areas of the two boxes, and combines them with intersection over union (IoU)-series losses in a weighted manner, effectively improving the localization accuracy of tiny targets. Finally, distribution-shift convolution (DSConv) is used in the neck to replace standard convolution and reduce computational cost.

These methods have significantly promoted the development of remote sensing object detection, but most of them are still built on convolution-based local modeling frameworks and remain limited in their use of long-range contextual relations and cross-layer semantic collaboration. In particular, in scenes where dense small targets and complex backgrounds coexist, the collaborative mechanism between shallow details and deep semantics still needs further improvement.

DETR is the first to systematically introduce Transformers into object detection tasks, achieving a truly end-to-end detection paradigm through set prediction. By using global self-attention to establish relationships between image features and object queries, it effectively avoids handcrafted components such as anchor design and nonmaximum suppression. However, its training converges slowly, and its adaptability to multiscale targets is relatively insufficient. To address this issue, Deformable DETR performs feature aggregation on a small number of key sampling points through deformable attention, significantly improving training efficiency and enhancing multiscale modeling capability. Since then, a series of improved DETR methods have continuously enhanced detection performance and training stability from the perspectives of query design, matching strategy, denoising training, and feature interaction.

For real-time scenarios, RT-DETR further optimizes the encoder, decoder, and query-selection mechanism, achieving high inference efficiency while maintaining detection accuracy; RT-DETRv2 further improves baseline performance through a series of training and structural enhancements. Owing to the advantages of global modeling and end-to-end prediction, Transformer detectors provide new ideas for remote sensing object detection. However, existing RT-DETR series models still mainly rely on conventional multiscale semantic feature inputs and make insufficient use of the shallow high-resolution detail information that is extremely important in remote sensing images. Meanwhile, semantic enhancement for low-level features is often coarse and still fails to fully consider the prominent directional structures, large contextual spans, and strong position sensitivity in remote sensing scenes. Therefore, how to more effectively exploit shallow detail information and enhance low-level semantic representation while preserving the real-time and end-to-end advantages of RT-DETR remains a problem worthy of further study.

In recent years, as RT-DETR/RT-DETRv2 has gradually demonstrated its advantages in real-time object detection, an increasing number of studies have introduced Transformer-based detection frameworks into remote sensing and UAV scenarios, with improvements targeting small objects, dense targets, drastic scale variations, and complex backgrounds. For example, Freq-DETR enhances detail preservation and foreground-background discrimination through frequency-enhanced convolution and high-low frequency interaction. PSWP-DETR focuses on improving deformation modeling and multiscale integration in complex scenes. CAEM-DETR improves small object detection performance by incorporating contrastive attention and multidomain fidelity fusion. CSFPR-RTDETR enhances RT-DETR for UAV scenarios from the perspective of cross-spatial–frequency feature modeling and positional relationship modeling. Meanwhile, SED-DETR [45] improves the adaptability to multiscale remote sensing targets by jointly modeling scale priors and shape priors through scale aggregation and deformable dilated sampling. Position-DETR [46] enhances small-object focus and convergence efficiency through hierarchical foreground selection, fine-grained foreground enhancement, and progressive attention optimization. SAFF-DETR [47] strengthens unified modeling for targets of different scales by introducing shallow frequency-representation fusion and cross-layer spatial–frequency/channel–frequency interactions. A scale-aware multidomain DETR [48] improves feature representation for small objects and complex backgrounds through dynamic scale perception, multidomain pyramid enhancement, and background suppression. OWRT-DETR [49] further improves real-time detection performance in UAV scenarios by adopting cross-scale feature pyramid interaction, multiscale perception fusion, and a small-object enhancement module. These studies indicate that existing Transformer/RT-DETR improvements for remote sensing mainly focus on frequency-information enhancement, scale-adaptive modeling, multiscale fusion, foreground selection and attention design, as well as positional relationship modeling.

However, although these methods improve detection performance from the perspectives of frequency information, scale modeling, foreground selection, multidomain enhancement, and cross-scale interaction, most of them still treat shallow features as a unified shared input and rarely explicitly consider the potential functional conflict between detail preservation and semantic transmission in shallow features. In addition, the enhancement of low-level semantic features in existing methods is still largely limited to general context compensation, frequency fusion, or attention-based recalibration, without being specifically designed for the prominent directional structures, drastic scale variations, and strong positional sensitivity in remote sensing scenes. Different from existing methods, the proposed DSH-DETR does not merely enhance a single-scale feature, frequency-domain information, or attention module. Instead, within the RT-DETRv2 framework, it simultaneously introduces detail-semantic decoupling, multiscale position-aware semantic enhancement, and high-resolution semantic reconstruction, thereby improving object detection performance in complex

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remote sensing scenes from both the perspective of shallow-feature functional specialization and low-level semantic compensation.

The above methods mainly improve detection performance from the perspectives of frequency-domain detail enhancement, multiscale context modeling, relationship modeling, or discriminative feature fusion, and have achieved promising results in complex remote sensing scenes. However, most of these methods still operate on shared shallow feature representations; that is, they further enhance representational capacity without changing how shallow features are used. For dense small-object detection in remote sensing images, although such strategies can improve feature quality to some extent, they do not explicitly address the potential representational conflict arising when shallow features simultaneously serve the two roles of detail preservation and semantic transmission. Meanwhile, they also lack more targeted enhancement mechanisms for issues such as weakened low-level semantics, prominent directional structures, and strong positional sensitivity.

Overall, existing remote sensing small-object detection methods mainly improve performance through frequency-domain detail preservation, multiscale feature fusion, texture and boundary enhancement, and classification-regression task decoupling. These methods alleviate problems such as information loss, scale variation, and interference from complex backgrounds from different perspectives, but most still focus on enhancing a particular type of information or optimizing a specific detection stage, rather than explicitly addressing the conflict between detail preservation and semantic transmission from the perspective of functional role assignment in shared shallow features. At the same time, for the problem that low-level feature semantics are prone to weakening, existing methods also lack specialized modeling mechanisms that take into account the prominent directional structures, severe scale variations, and strong spatial sensitivity characteristic of remote sensing scenes.

Motivated by this, this article further introduces detail-semantic decoupling and multiscale position-aware semantic enhancement into the RT-DETR framework to simultaneously alleviate two key problems: the conflict in shallow features and the insufficiency of low-level semantics. It should be noted that the "decoupling" referred to in this article does not mean high-frequency/low-frequency separation in the frequency-domain sense, nor does it refer to the decoupling of classification and regression tasks in the detection head; rather, it denotes an explicit functional separation of shared shallow features into a detail-preservation path and a semantic-modeling path.

B. Multiscale Feature Fusion

Multiscale feature fusion is an important way to improve small-object detection performance. FPN [21] enhances the semantic expressive ability of shallow features by passing high-level semantic information through a top-down path, while PANet [22] further introduces bottom-up path aggregation to strengthen information flow and feature complementarity across different scales. Subsequent studies have further improved multiscale modeling effects through bidirectional feature pyramids, cross-scale feature recalibration, and dynamic weighted fusion. For example, Tong and Wu [23] proposed DFLFFN, a deep feature learning and feature fusion network for small-object detection. First, a deep feature learning module (DFLM) is designed to activate multiscale receptive fields within a single layer using convolutions with different dilation rates, thereby capturing rich semantics and contextual information and enhancing the characterization of small targets in different scenes. Second, a feature fusion block (FFB) is proposed to concatenate and fuse deep semantic features enhanced by DFLM with shallow detail features, compensating for the insufficient semantics of shallow features and improving small-target classification and localization accuracy. Liu et al. [24] proposed MSCAF-Net, a multiscale context-aware feature learning framework for camouflaged object detection. It first adopts an improved pyramid vision transformer (PVTv2) as the backbone to effectively extract multiscale global contextual information. Second, it designs an enhanced receptive field module (ERF) that combines dilated convolution and asymmetric convolution to enrich multiscale contextual features at each scale. Then, a cross-scale feature fusion module (CSFF) is introduced to enable sufficient interaction among multiscale features through concatenation and conditional batch normalization, while a coarse localization map is used to modulate fused features and enhance scale diversity. In addition, inspired by the human visual system, a dense interactive decoder (DID) is designed to generate coarse localization maps based on multiscale features and guide subsequent precise segmentation.

Such methods play an important role in handling remote sensing targets with obvious scale variations, but if they rely only on direct concatenation or simple weighting, it remains difficult to simultaneously preserve shallow details and compensate for high-level semantics.

C. Attention Mechanism and Position-Aware Modeling

Attention mechanisms have been widely used to improve feature selection ability and target response strength. Channel attention can highlight key channels related to target discrimination, spatial attention helps strengthen responses in salient regions, and CA enhances position-information modeling under lightweight conditions by separately encoding horizontal and vertical information. For example, Li et al. [25] proposed DI-YOLOv5, an improved YOLOv5 algorithm based on dual wavelets for dense small-object detection. First, a dual-wavelet convolution module (DWCM) is designed, including DWT_Conv and IWT_Conv, which replace traditional downsampling and upsampling operations, respectively. DWT_Conv decomposes feature maps into low- and high-frequency subbands through discrete wavelet transform, preserving detail information while reducing resolution; IWT_Conv restores high-resolution feature maps through inverse wavelet transform, effectively reducing information loss caused by downsampling and checkerboard artifacts caused by upsampling. Second, a CSPCoA module is proposed, which introduces a parallel coordinate-attention path into the CSP structure. By encoding positional information along the horizontal, vertical, and channel directions, it

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captures long-range dependencies and precise positional information, thereby enhancing the localization ability for dense targets. Wang et al. [26] proposed LSOD-YOLO, a lightweight small-object detection algorithm improved from YOLOv8. First, a lightweight cross-layer output reconstruction module (LCOR) is designed. By removing the high-level feature layer (P5), which contributes less to object detection, and adding cross-layer connections, it fuses shallow and deep features, significantly reduces parameter count and computational complexity, and improves small-object detection ability. Second, a new spatial pyramid pooling layer (SPPFL) is constructed by incorporating large separable kernel attention (LSKA) into the SPPF structure. It captures long-range dependencies and multiscale contextual information through depthwise separable dilated convolution, thereby improving global feature representation. Then, a C2f-N module based on the normalized attention mechanism (NAM) is proposed, which suppresses nonsalient features and highlights key features through channel and spatial normalization strategies, thereby strengthening the model's ability to extract small-target features under complex backgrounds. Finally, a lightweight dynamic upsampler, dysample, is introduced. It dynamically generates sampling-point offsets to replace nearest-neighbor interpolation, reducing pixel distortion and information loss and improving upsampling accuracy. Liu et al. [27] proposed UCIW-YOLO, a lightweight high-accuracy model for farmland obstacle detection, improved from YOLOv5s. First, a new dataset containing nine types of farmland obstacles, including soil piles, people, utility poles, transmission towers, trees, tractors, grain trucks, harvesters, and irrigation facilities, is constructed. It covers different seasons, weather conditions, and crop backgrounds, and is expanded to 22 093 images through multiple data augmentation methods. Second, a lightweight C3UIB module is designed, using the Extra DW module from MobileNetV4 in the neck network to reduce parameter count and computational complexity while maintaining feature-fusion capability. Then, a C3UIBCA module is proposed, which introduces the CA mechanism into C3UIB and deploys it in the backbone network. By encoding positional information in horizontal and vertical directions, it strengthens the spatial localization ability for elongated and small-scale obstacles. Finally, an Inner-WIoU loss function is proposed. By combining the dynamic nonmonotonic focusing mechanism of WIoUv3 with the auxiliary bounding-box acceleration strategy of Inner-IoU, it improves bounding-box regression accuracy.

For remote sensing images, directional structures, target arrangement patterns, and complex-background suppression are also crucial. Therefore, combining multiscale feature fusion with position-sensitive attention mechanisms has become an important direction for improving remote sensing object detection performance.

Compared with existing methods, the method proposed in this article differs in the following aspects.

1) In the semantic branch, an MPSEM is constructed. Unlike existing methods that mainly rely on single-scale convolution, channel attention, or spatial attention for feature enhancement, this module integrates strip convolution, dilated convolution, adaptive semantic interactive fusion, and coordinate-attention calibration to jointly model directional structures, scale variations, long-range dependencies, and spatial positional information. As a result, it is better suited to the complex feature distributions of remote sensing images, which are typically characterized by large scale variations, dense object arrangements, and strong background interference.
2) At the feature output stage, an HRSM is incorporated. Different from existing multiscale detection frameworks that rely solely on the original pyramid features for prediction, high-resolution semantic representations are further reconstructed while preserving the original semantic hierarchy. These reconstructed features, together with the original multiscale features, form a four-scale detection architecture, thereby improving the perception of small targets, weak targets, and targets embedded in complex backgrounds.

Overall, existing Transformer/RT-DETR-based remote sensing detection methods mainly improve detection performance from the perspectives of frequency-domain enhancement, multiscale fusion, context modeling, and positional relationship modeling. However, most of these methods still operate on shared shallow feature representations and do not explicitly address the functional conflict between detail preservation and semantic transmission in shallow features. Different from these methods, the core of this work does not lie in further stacking enhancement modules on shared shallow features, but in re-examining how shallow features are utilized at the encoding stage. Specifically, starting from the problems of small-object detail loss, insufficient shallow semantic representation, and functional mixing in shared features in remote sensing scenes, this article constructs a detail-semantic dual-branch diversion mechanism, allowing shallow features to be specialized along a detail-preservation path and a semantic-modeling path, respectively. Subsequently, an MPSEM is introduced into the semantic path. Through strip convolution, dilated convolution, adaptive semantic interactive fusion, and CA calibration, the module jointly models directional structures, scale variation, long-range dependencies, and spatial positional information. Finally, an HRSM is employed to generate additional high-resolution compensatory features, thereby improving the completeness and localization accuracy of small-object detection. Therefore, the essential difference between this work and existing Transformer/RT-DETR-based methods is that the latter mostly enhance shared shallow features directly, whereas this article first alleviates shallow representation conflicts through explicit functional branching and then performs targeted enhancement for weakened low-level semantics, thereby forming an improvement path better suited to remote sensing small-object detection.

The remainder of this article is organized as follows. Section II reviews related work on remote sensing object detection and Transformer-based detectors. Section III presents the proposed DSH-DETR in detail, including the detail-semantic decoupling module, the MPSEM, and the high-resolution semantic reconstruction strategy. Section IV describes the experimental settings and reports comparative results, abla-

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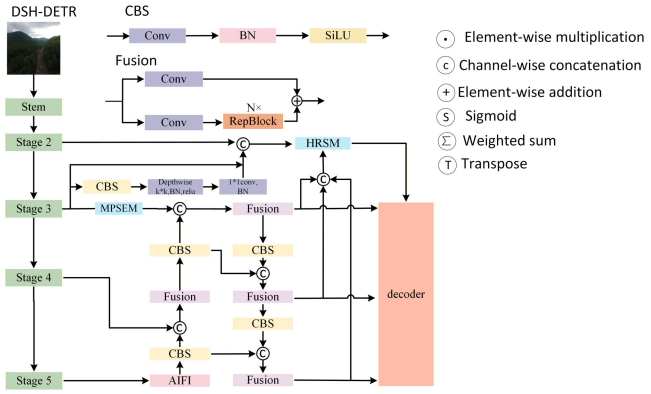


Fig. 1. Overall framework of DSH-DETR.

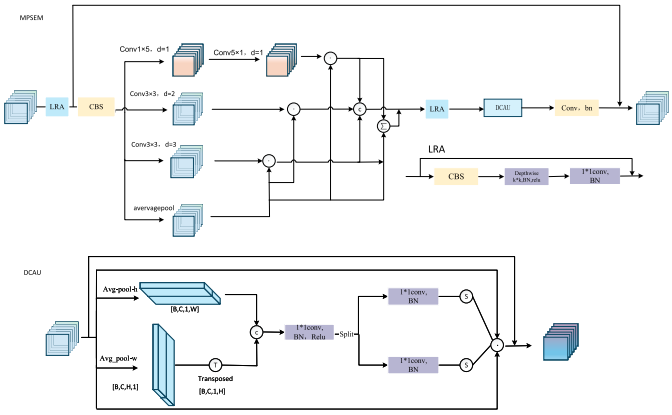


Fig. 2. Multiscale position-aware semantic enhancement module.

tion studies, visualization analyses, and complexity evaluation on the VisDrone2019-DET, RSOD, USOD, and RS-STOD datasets. Section V concludes this article and discusses future work.

III. Method

A. Overall Framework

To address the issues of large-scale variation, dense small targets, complex backgrounds, and the progressive loss of shallow detail information caused by repeated downsampling in remote sensing images, a remote sensing object detection framework based on detail-semantic decoupling and multiscale coordinate-guided semantic enhancement is developed upon RT-DETRv2, denoted as DSH-DETR. The overall framework is shown in Fig. 1. The input image first passes through a ResNet-50vd backbone to extract four levels of features, denoted as C_2, C_3, C_4, and C_5, respectively. The backbone feature extraction process can be expressed as

$$\{C_2, C_3, C_4, C_5\} = \text{Backbone}\,(I) \tag{1}$$

where I denotes the input image, and Backbone$(\cdot)$ denotes the feature extraction process of the backbone network. In the current framework, shallow features are no longer directly shared with all subsequent paths. Instead, they are first split into two branches, one for detail and the other for semantics, at the encoding stage. As shown in the figure, these two branches are denoted as the "detail path" and the "semantic path." Specifically, C_3 is divided into the detail feature D_2 and the semantic feature S_2. The former is used for high-resolution detail compensation, while the latter is used for subsequent low-level semantic enhancement. This process can be expressed as

$$D_2 = R_d\,(C_3) \tag{2}$$

where $R_d(\cdot)$ denotes the mapping process of the detail path. Through this dual-branch splitting strategy, detail information and semantic information in shallow features can be optimized along different paths, thereby alleviating the representation conflicts caused by shared features.

Subsequently, the basic semantic feature C_3 is fed into the MPSEM to obtain the enhanced semantic feature S_2, namely

$$S_2 = \text{MPSEM}\,(C_3)\,. \tag{3}$$

The enhanced semantic feature S_2, together with the high-level features C_4 and C_5, is then input into the encoder. After deepest-layer feature encoding and FPN-PAN multiscale aggregation, three semantic feature maps F_3, F_4, and F_5 are generated, namely

$$\{F_3, F_4, F_5\} = \text{Encoder}\,(S_2, C_4, C_5)\,. \tag{4}$$

Meanwhile, the shallow high-resolution feature C_2, the detail feature D_2, and the encoder-output semantic features F_3, F_4, and F_5 are jointly fed into the HRSM to generate an additional high-resolution compensatory feature SS, namely

$$SS = \text{HRSM}\big(C_2, D_2, F_3, F_4, F_5\big). \tag{5}$$

Finally, the decoder receives four levels of features, $\{SS, F_3, F_4, F_5\}$, for end-to-end target classification and bounding-box regression, forming a four-scale detection framework. This process can be expressed as

$$Y = \text{Decoder}\big(SS, F_3, F_4, F_5\big) \tag{6}$$

where Y denotes the final detection result. The core idea of the proposed method is to explicitly separate detailed information from semantic information at the shallow stage, then enhance context modeling and position-awareness capability in the semantic path, and finally provide the decoder with high-resolution feature representations that are more suitable for small-object detection through the high-resolution semantic reconstruction branch.

B. Multiscale Position-Aware Semantic Enhancement Module

To enhance the modeling capacity of low-level semantic features for directional structures, long-range dependencies, and complex background context, an MPSEM is constructed, the architecture of which is illustrated in Fig. 2. Taking the basic semantic feature C_3 as input, the module first preserves a stable basic semantic pathway and builds a multibranch semantic enhancement structure on top of it. To ensure the stability of the enhancement process, the input feature is first transformed through basic adaptation and preprocessing mappings to obtain the intermediate features B and X_0, namely

$$B = \text{LRA}\,(C_3)\,, \qquad X_0 = \text{CBS}\,(B)\,. \tag{7}$$

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On this basis, the module constructs one strip branch and two dilated branches in parallel to extract semantic information with heterogeneous receptive fields. The strip branch adopts cascaded $1 \times k$ and $k \times 1$ depthwise convolutions to model horizontal and vertical directional structural information, while the two dilated branches employ 3×3 depthwise convolutions with different dilation rates to enlarge the effective receptive field and supplement broader contextual responses. The extraction process of the three semantic branches can be expressed as

$$U_1 = \Phi_{\text{strip}}(X_0), \qquad U_2 = \Phi_{\text{dil},2}(X_0), \qquad U_3 = \Phi_{\text{dil},3}(X_0) \tag{8}$$

where U_1, U_2, and U_3 denote the output features of the strip branch, the first dilated branch, and the second dilated branch, respectively. Given that the contributions of different branches are not equally important across scenarios, a global gated weighting mechanism is further incorporated. This mechanism first extracts a global semantic representation of the input features via global pooling, then generates branch-specific weights through a mapping function, and finally adaptively recalibrates the responses of all branches after Softmax normalization. The process can be written as

$$w = \text{Softmax}\left(\text{AvgPool}(X_0)\right), \qquad \hat{U}_i = w_i \cdot U_i \tag{9}$$

$$U_{\text{id}} = \sum_{i=1}^{3} \hat{U}_i, \qquad i = 1, 2, 3 \tag{10}$$

where $w = [w_1, w_2, w_3]$ denotes the normalized weights of the three branches, $\hat{U}_i$ denotes the weighted branch features, and U_{id} denotes the main semantic aggregation result after weighted summation. This design dynamically adjusts the contribution ratios of different semantic branches according to the input scene content. Although weighted summation is capable of preserving the dominant semantic responses, it remains insufficient for fully exploiting the complementary information across branches. Therefore, a residual semantic fusion mechanism is further incorporated, where the weighted multibranch features are concatenated along the channel dimension and passed through a lightweight semantic fusion mapping to strengthen interbranch interaction. The process can be expressed as

$$U_{\text{cat}} = \text{Concat}\left(\hat{U}_1, \hat{U}_2, \hat{U}_3\right) \tag{11}$$

$$F = U_{\text{id}} + \alpha_f \cdot U_{\text{cat}} \tag{12}$$

where Concat$(\cdot)$ denotes channel concatenation, α_f is a learnable fusion coefficient, and F denotes the fused semantic feature. Through this design, multibranch information not only preserves the weighted contribution of each branch, but also enables more sufficient interaction during fusion.

To further enhance the sensitivity of semantic features to spatial positional information, a coordinate-attention calibration mechanism is introduced after semantic fusion. This mechanism aggregates global information along the height and width directions, explicitly models long-range dependencies along both directions, and generates directional attention weights accordingly. The process can be expressed as

$$Z_h = P_h(F), \qquad Z_w = P_w(F) \tag{13}$$

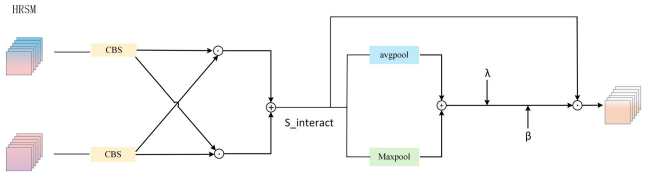


Fig. 3. High-resolution semantic reconstruction module.

$$Q = f\left(\text{Concat}\left(Z_h, Z_w^{\top}\right)\right) \tag{14}$$

$$A_h = \text{sigmoid}\left(f_h(Q_h)\right), \qquad A_w = \text{sigmoid}\left(f_w(Q_w)\right) \tag{15}$$

$$R = F \odot A_h \odot A_w \tag{16}$$

where $P_h(\cdot)$ and $P_w(\cdot)$ denote aggregation operations along the height and width directions, respectively; $f(\cdot)$ denotes shared mapping; $f_h(\cdot)$ and $f_w(\cdot)$ denote the attention-generation functions for the two directions; and R denotes the semantic feature calibrated by CA.

Finally, MPSEM outputs the enhanced semantic feature S_2 through residual injection, namely

$$S_2 = B + \alpha_r \cdot R \tag{17}$$

where α_r is a learnable residual coefficient. Through the progressive enhancement process of "basic semantic preservation—multiscale branch extraction—adaptive branch weighting—residual-style semantic fusion—coordinate-attention calibration," MPSEM effectively improves the context-modeling ability and spatial position-awareness of low-level semantic features for small targets in complex remote sensing scenes, thereby constructing stronger low-level semantic inputs for the subsequent encoder.

C. High-Resolution Semantic Reconstruction Module

To alleviate the conflict between the insufficient resolution of high-level features and the limited semantic representation capability of shallow features, an HRSM is designed, as illustrated in Fig. 3. Built upon the high-resolution detail information from shallow features, this module further incorporates multiscale high-level semantic features from the encoder output. Through cross-layer fusion and attention-based recalibration, an additional high-resolution compensatory feature, denoted as SS, is generated, thereby improving the localization and recognition of small targets in complex scenes.

First, the module projects the shallow feature C_2 and the detail feature D_2 into a unified channel space, and aligns D_2 to the target resolution to obtain the shallow spatial feature S_1 and the detail compensatory feature D_2', namely

$$S_1 = \text{Proj}_1(C_2), \qquad D_2' = \text{Up}\left(\text{Proj}_2(D_2)\right) \tag{18}$$

where $\text{Proj}_1(\cdot)$ and $\text{Proj}_2(\cdot)$ denote linear projection, and Up$(\cdot)$ denotes upsampling.

Subsequently, to fully fuse high-level semantic information at different scales, the module upsamples F_3, F_4, and F_5 to the same resolution and constructs the aggregated semantic feature S_{pan} through adaptive weighting. The specific process is as follows:

$$F_3' = \text{Up}(F_3), \qquad F_4' = \text{Up}(F_4), \qquad F_5' = \text{Up}(F_5) \tag{19}$$

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$$[a_3, a_4, a_5] = \text{Softmax}\left(\text{Agg}\left(\text{Concat}\left(F_3', F_4', F_5'\right)\right)\right) \tag{20}$$

$$S_{\text{pan}} = a_3 \cdot F_3' + a_4 \cdot F_4' + a_5 \cdot F_5' \tag{21}$$

where a_3, a_4, and a_5 denote the normalized weights of the semantic features at the three scales, Agg(·) denotes the weight-generation function, and S_{pan} denotes the aggregated high-level semantic feature. This design enables the model to adaptively select more effective scale-specific semantic responses according to the input scene.

For shallow spatial detail modeling, S_1 and D_2' are concatenated along the channel dimension, and a high-resolution spatial representation, denoted as S_{spa}, is generated through convolutional mapping, i.e.,

$$S_{\text{spa}} = \text{Conv}\left(\text{Concat}\left(S_1, D_2'\right)\right). \tag{22}$$

After obtaining the shallow spatial features and high-level semantic features, HRSM further constructs a bidirectional spatial-semantic interactive representation, so that shallow details are enhanced under the guidance of high-level semantics, while high-level semantics also receive finer localization support. This process can be expressed as

$$S_{\text{interact}} = S_{\text{spa}} \odot S_{\text{pan}} + S_{\text{pan}} \odot S_{\text{spa}} \tag{23}$$

where S_{interact} denotes the fused feature after spatial-semantic interaction.

To further highlight the response channels that are more critical for small-target discrimination, a parallel channel recalibration mechanism is introduced based on the interactive features, where channel descriptors are extracted from two branches via global average pooling and global max pooling, respectively. The process can be expressed as

$$A_{\text{avg}} = g_{\text{avg}}\left(\text{GAP}\left(S_{\text{interact}}\right)\right), \qquad A_{\text{max}} = g_{\text{max}}\left(\text{GMP}\left(S_{\text{interact}}\right)\right) \tag{24}$$

$$A = \lambda \cdot \left(A_{\text{avg}} + A_{\text{max}}\right) + \beta \tag{25}$$

where GAP(·) and GMP(·) denote global average pooling and global max pooling, respectively; $g_{\text{avg}}(\cdot)$ and $g_{\text{max}}(\cdot)$ denote the corresponding channel mapping functions; λ and β are learnable scaling parameters; and A denotes the final channel recalibration weights.

Finally, the module outputs the high-resolution compensatory feature SS, namely

$$SS = S_{\text{interact}} \odot A. \tag{26}$$

Compared with directly using shallow features or simple multiscale concatenation, HRSM can effectively integrate shallow detail information and high-level semantic context within a unified resolution space, and enhance responses related to small targets through adaptive weight allocation and channel recalibration. The final high-resolution feature SS is sent together with F_3, F_4, and F_5 into the decoder for joint detection, thereby providing more sufficient feature support for dense small-target scenes.

In summary, through three key components, namely, detail-semantic dual-branch splitting, MPSEM-based multiscale semantic enhancement, and HRSM-based high-resolution semantic reconstruction, the proposed method realizes collaborative modeling of shallow detail preservation, low-level semantic strengthening, and high-resolution feature compensation, thereby effectively improving detection performance in complex remote sensing scenes, especially those dominated by small targets.

D. Motivation for the Split and Its Implementation Basis

Considering that C3 still retains relatively high spatial resolution, its features simultaneously contain local texture details and preliminary semantic information. If these two types of information are uniformly enhanced within a single pathway, mutual interference is likely to arise between detail response and semantic aggregation. Therefore, instead of explicitly splitting the C3 features along the channel dimension, this article adopts a homology-based dual-branch decoupling strategy: the same C3 feature is separately fed into a detail-adaptive branch and a semantic-adaptive branch to generate D2 and S2, respectively. The former is passed through a local residual mapping with a relatively large receptive field and then sent to SIEU to strengthen texture and edge information; the latter is enhanced by MPSEM and then fed into the FPN/PAN for semantic fusion with deeper features. This design avoids manually specifying the channel allocation ratio, allowing the division of labor between detail and semantics to be adaptively learned by the network during training. It should be noted that this decoupling does not aim at a strict hard separation in the frequency domain; rather, it reduces the competition between different types of information within the same pathway through branch bias. Its effectiveness will be further validated in G. Rationality Analysis of D2/S2 Decoupling through spectral analysis and activation visualization.

E. Loss Function

Following RT-DETRv2, the proposed method does not introduce any additional handcrafted loss term. To ensure that the performance gains mainly come from the proposed architectural improvements, the same detection objective as the baseline is adopted. Specifically, Hungarian matching is first used to establish one-to-one assignment between predictions and ground truths. The overall detection loss is defined as

$$\mathcal{L} = \lambda_{vfl}\mathcal{L}_{vfl} + \lambda_{bbox}\mathcal{L}_{bbox} + \lambda_{giou}\mathcal{L}_{giou}$$

where $\mathcal{L}_{vfl}$ denotes the varifocal classification loss, $\mathcal{L}_{bbox}$ denotes the L1 bounding box regression loss, and $\mathcal{L}_{giou}$ denotes the generalized IoU loss. In our implementation, $\lambda_{vfl} = 1$, $\lambda_{bbox} = 5$, and $\lambda_{giou} = 2$. The same supervision is also applied to auxiliary outputs during training.

IV. Experiments

A. Datasets

Considering that objects in remote sensing images are generally characterized by small scales, dense distributions, complex backgrounds, and pronounced scale variations, four datasets, namely, VisDrone2019-DET [28], RSOD [29], USOD [30], and RS-STOD [43], are selected to evaluate the effectiveness of the method. Among them, VisDrone2019-DET serves as the main experimental dataset and is used

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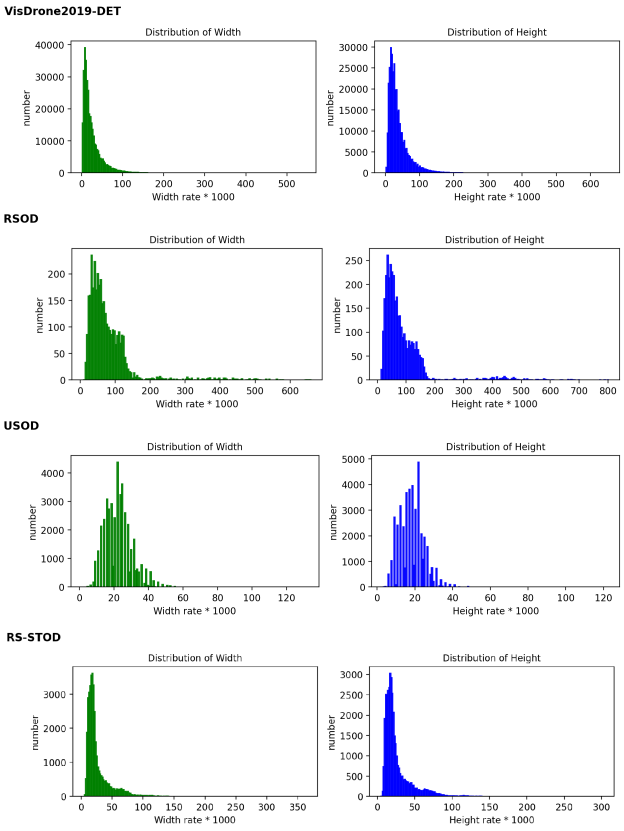


Fig. 4. Statistics of normalized target width and normalized target height distributions.

to evaluate the model's small-object detection capability in complex UAV scenes; RSOD is used to verify the model's generalization performance in generic remote sensing object detection tasks; and USOD is further used to evaluate the detection performance of the model under conditions involving extremely small targets, low illumination, occlusion, and degradation. RS-STOD is further used to evaluate the model's detection performance in challenging remote sensing scenes characterized by super-tiny objects, dense distributions, and complex background interference. Fig. 4 presents the statistics of normalized target width and normalized target height in the four datasets, where width and height are obtained by dividing the target-box width and height by the corresponding image width and height, respectively, and then multiplying by 1000. It can be seen that targets in all four datasets are generally concentrated in the small-scale region, indicating that the experimental tasks in this article exhibit significant small-object detection characteristics. Among them, the distributions of VisDrone2019-DET and USOD are closer to the lower value region, showing stronger small-target challenges, whereas RSOD has a relatively larger scale span and is therefore more suitable for evaluating the model's multiscale adaptability.

1) VisDrone2019-DET: VisDrone2019-DET is a widely used benchmark dataset for object detection in the UAV vision field, containing 10 209 high-resolution static images. The dataset covers a variety of real UAV scenes, including urban roads, residential areas, squares, and parking lots, and contains ten target categories: pedestrian, people, bicycle, car, van, truck, tricycle, awning-tricycle, bus, and motor. Following the commonly used setting of this dataset, 6471 images are used as the training set and 548 images are used as the validation set. As can be seen from the last row of Fig. 4, the target width and height distributions in VisDrone2019-DET are highly concentrated in the low-value region and exhibit a clear long-tail characteristic, indicating that the dataset contains both a large number of extremely small targets and a certain number of medium-scale targets. Therefore, it is well suited for evaluating the model's joint modeling capability for dense small targets and multiscale targets.

2) RSOD: RSOD is a typical remote sensing object detection dataset containing 976 images and covering four categories: aircraft, oil tank, playground, and overpass. Compared with UAV-scene datasets, RSOD is more oriented toward the traditional top-down perspective of remote sensing, and its target distribution exhibits more obvious scale variation. As can be seen from the first row of Fig. 4, although the target width and height in RSOD are also mainly concentrated in a relatively small interval, the overall distribution is more dispersed and the long tail on the right is more obvious, indicating that the dataset contains not only many small targets, but also a certain proportion of medium- and large-scale targets. Therefore, RSOD is more suitable for validating the model's generalization performance and robustness in multiscale remote sensing scenes.

3) USOD: USOD is a dataset designed for remote sensing tiny-object detection and is mainly used for small-vehicle detection research. It contains 3000 images, with small targets accounting for more than 99.9%. In addition, USOD includes many complex scenes involving low illumination and shadow occlusion, and also provides test sets under degraded conditions such as blur, Gaussian noise, stripe noise, and haze, which can be used to evaluate the robustness of detection models in complex imaging environments. As can be seen from the second row of Fig. 4, the target width and height distributions in USOD are the most concentrated, with the main peak clearly clustered within a lower numerical range, indicating that the dataset exhibits a stronger "extremely small target dominance" characteristic. Therefore, USOD can effectively verify the performance advantages of the proposed method in preserving high-resolution details, enhancing weak-target responses, and suppressing complex backgrounds.

These four datasets provide a relatively comprehensive validation of the proposed method from three perspectives: complex UAV scenes, small-target-dominated scenes, and multiscale remote sensing scenes. Among them, VisDrone2019-DET focuses on evaluating dense small-object detection capability in complex scenes, USOD emphasizes the model's adaptability to extremely small targets and degraded scenes, and RSOD is used to measure the model's generalization ability under a broader scale distribution of remote sensing targets.

4) RS-STOD: RS-STOD is a dataset designed for remote sensing small object detection and is mainly used to evaluate the capability of detection models to recognize small-scale targets in complex remote sensing scenes. The dataset contains

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TABLE I
HARDWARE AND SOFTWARE ENVIRONMENTS

Hardware Environment	
CPU	2 × Intel Xeon Gold 5218R @ 2.10 GHz
RAM	251 GiB
Video memory	48 GB × 8
GPU	NVIDIA A40 × 8
Software Environment	
OS	Linux
CUDA	11.8
Python	3.8
Torch	2.10

1852 training images and 463 validation images, covering five target categories. Compared with general remote sensing object detection datasets, objects in RS-STOD are smaller and more densely distributed, while the scene backgrounds are more complex, often accompanied by low contrast, background texture interference, dense target arrangements, and significant scale variation. These characteristics place higher demands on a model's fine-grained representation ability and robustness to interference. As shown by the statistical results in the figure, the target widths and heights in RS-STOD are mainly concentrated in relatively low value ranges, indicating that the dataset is characterized by a pronounced dominance of small objects. Meanwhile, its distribution tail still retains a certain scale span, suggesting that the dataset also contains a small number of medium-scale targets. Therefore, RS-STOD can effectively validate not only a model's detection capability for extremely small and densely distributed objects, but also its robustness and multiscale adaptability under complex background conditions.

These four datasets provide a relatively comprehensive validation of the proposed method from three perspectives: complex UAV scenes, small-object-dominated scenes, and multiscale remote sensing scenes. Among them, VisDrone2019-DET mainly evaluates dense small-object detection performance in complex scenes; USOD emphasizes the model's adaptability to extremely small objects and degraded scenarios; and RS-STOD is primarily used to assess the detection performance of the proposed method in remote sensing scenes dominated by small objects. This dataset is particularly effective for evaluating the model's ability to preserve weak detail information, as well as its perception and localization capability for small targets. RSOD, in contrast, is used to measure the model's generalization ability over a broader scale distribution of remote sensing targets.

B. Implementation Details

To minimize result fluctuations caused by differences in experimental platforms and to ensure the stability and reproducibility of the experimental results, all experiments were conducted on the same server. All YOLO- and DETR-series models were implemented using their official codebases.

Both the proposed method and the comparison models were initialized with their corresponding pretrained weights, followed by fine-tuning. The CUDA version was 11.8, and the random seed for the proposed method was set to 42. The data were organized in COCO format, and CocoEvaluator was used for performance evaluation. The input size in both training and validation was set to 640 × 640. During training, data augmentation strategies including RandomPhotometricDistort, RandomZoomOut, RandomIoUCrop, RandomHorizontalFlip, and Resize were adopted. The batch size in both training and validation was set to 4, and num_workers was set to 4. AdamW was used as the optimizer. The hardware and software environments used in the experiments are summarized in Table I.

C. Evaluation Metrics

AP, which is commonly used in object detection tasks, and inference speed measured by frames per second (FPS) are adopted as the primary evaluation metrics to comprehensively assess the detection performance and efficiency of the model. First, AP is adopted as the primary metric for accuracy evaluation. Specifically, AP denotes the average detection precision computed over IoU thresholds ranging from 0.50 to 0.95 with a step size of 0.05, and is used to comprehensively measure the overall detection capability of the model. AP_{50} and AP_{75} denote the AP at IoU = 0.50 and IoU = 0.75, respectively, reflecting the detection performance of the model under relatively loose and more stringent localization criteria. In addition, detection performance across different object scales is further evaluated using AP_S, AP_M, and AP_L, which correspond to the AP for small, medium, and large objects, respectively. Since the focus is placed on small-object detection in remote sensing images, AP_S is regarded as one of the key metrics for analysis.

To further assess the practical applicability of the model, FPS is adopted to measure inference speed, i.e., the number of images processed per second, thereby reflecting the real-time detection capability of the model. Meanwhile, Params is used to denote the number of model parameters and to characterize model complexity. By jointly considering AP, FPS, and Params, the performance and efficiency of the method in remote sensing small-object detection can be evaluated more comprehensively.

D. Comparison Algorithms

To comprehensively evaluate the effectiveness of the proposed method, representative state-of-the-art object detection methods are selected as baselines across different datasets, covering single-stage convolutional detectors, DETR-series Transformer detectors, and improved small-object detection methods for remote sensing or UAV scenarios. The YOLOv10, YOLOv11, and YOLOv12 series belong to typical one-stage object detection methods that balance detection accuracy and inference speed, and are commonly used comparison baselines in current real-time object detection tasks.

Anchor-DETR [34], DAB-DETR [35], and DN-DETR [36] are representative DETR-series methods that improve the

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TABLE II
COMPARATIVE ANALYSIS ON VISDRONE2019-DET

Method	backbone	AP^{val}	AP^{val}_{50}	AP^{val}_{75}	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}	FPS	Params (M)
yolov12x	–	27.1	44.1	27.7	17.0	40.2	51.8	24.5	60.3
yolov12l	–	25.5	41.8	26.3	15.6	38.3	51.4	17.3	26.8
yolov11x	–	26.6	43.8	27.0	16.8	39.5	52.8	35.8	56.8
yolov11m	–	24.8	41.1	25.2	14.9	37.5	50.5	60.9	20.0
yolov10m	–	23.0	38.8	23.2	13.4	35.5	41.8	50.6	16.4
Yolov10l	–	24.8	40.9	25.3	14.9	37.3	47.3	41.6	25.7
Anchor-DETR	Resnet50	21.8	39.9	20.9	12.9	33.0	47.1	10.4	37.0
DAB-DETR	Resnet50	18.6	36.3	16.9	9.9	28.6	45.2	19.2	43.6
DN-DETR	Resnet50	20.3	38.3	18.9	11.1	31.2	48.4	9.9	43.6
RT-DETR	Resnet50	30.5	51.0	30.8	21.2	42.1	55.5	32.4	49.8
PSWP-DETR	Resnet50	32.5	54.1	32.5	22.3	45.4	64.1	32.8	98.8
CAEM-DETR	GRACE-Net	30.8	50.2	31.8	23.0	40.7	42.5	65.0	19.9
CSFPR-RTDETR	CSFH	30.4	49.8	31.0	22.8	40.3	49.1	21.5	14.0
ours	Resnet50	33.7	55.0	34.4	24.9	45.1	53.2	16.9	48.9

TABLE III
COMPARATIVE ANALYSIS ON THE RSOD DATASET

Datasets	Method	$AP^{val}_{0.50:0.95}$	$AP^{val}_{0.50}$	$AP^{val}_{0.75}$	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}	FPS	Params (M)
RSOD	yolov12n	63.8	93.6	67.5	25.0	65.1	70.1	38.2	2.5
	yolov12l	68.7	92.5	77.7	33.5	72.1	73.6	21.5	26.8
	yolov11x	69.3	93.3	76.9	34.1	71.8	74.4	17.0	56.8
	yolov11m	67.9	92.2	77.1	34.6	70.6	73.5	30.7	20.0
	yolov10m	66.8	91.9	75.6	29.4	70.6	71.7	27.0	16.4
	Yolov10n	64.4	91.3	70.8	25.5	63.8	71.3	38.9	2.6
	AB-Net	–	94.1	–	–	–	–	–	42.8
	SAPF	–	90.4	68.8	–	–	–	–	37.3
	YOLO-SPCI	66.3	92.4	–	–	–	–	–	3.09
	RT-DETR	69.6	94.3	80.0	37.1	73.2	74.5	26.9	49.8
	ours	70.3	94.4	80.9	38.9	72.7	75.3	12.5	48.9

TABLE IV
COMPARATIVE ANALYSIS ON THE USOD DATASET

Datasets	Method	$AP^{val}_{0.50:0.95}$	$AP^{val}_{0.50}$	$AP^{val}_{0.75}$	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}	FPS	Params (M)
USOD	Yolo12x	35.6	91.5	15.1	35.8	34.3	-1	30.7	60.3
	Yolo12l	34.9	90.4	14.8	34.9	45.1	-1	30.2	26.8
	Yolo11x	35.8	91.1	15.8	35.8	41.6	-1	41.9	56.8
	Yolo11m	35.4	90.6	14.6	35.4	37.4	-1	52.7	20.0
	Yolo10m	34.9	89.0	14.6	34.9	39.4	-1	48.4	16.4
	Yolo10l	35.9	90.1	16.3	35.9	42.2	-1	42.4	25.7
	RT-DETR	36.7	91.9	16.5	36.7	45.7	-1	24.6	49.8
	FFCA-YOLO	33.3	88.6	13.4	33.4	15.6	-1	82.1	5.04
	ours	37.7	92.9	17.8	37.6	47.3	-1	6.5	48.9

TABLE V
COMPARISON RESULTS ON THE RS-STOD DATASET

Datasets	Method	$AP^{val}_{50:95}$	$AP^{val}_{0.50}$	$AP^{val}_{0.75}$	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}	FPS	Params (M)
RS-STOD	Anchor-DETR	34.6	61.4	34.9	32.1	62.1	-1	16.9	37.0
	DAB-DETR	26.2	49.1	26.0	23.0	57.2	-1	21.0	43.6
	DN-DETR	35.7	60.8	36.5	33.6	63.4	-1	16.4	43.6
	RT-DETR	49.0	73.2	54.3	47.2	69.1	-1	29.2	49.8
	FBV-Fusion	34.5	63.0	33.5	32.5	46.8	-1	37.2	91.6
	DCFL+RPUN-L	44.9	70.8	49.1	-	-	-	-	-
	ours	52.6	76.2	59.5	51.2	69.6	-1	9.72	48.9

performance of Transformer detectors from perspectives such as query design and training optimization.

RT-DETR [16] is a Transformer detector that balances real-time performance and detection accuracy, and it is also the basic framework on which the proposed method is built.

PSWP-DETR [31], CAEM-DETR [32], and CSFPR-RTDETR [33] are improved methods proposed for UAV or complex-scene detection tasks and are representative to some extent in multiscale modeling and small-object detection.

AB-Net [37], SAPF [38], YOLO-SPCI [39], and FFCA-YOLO [40] are detection methods designed for remote sensing or small-object scenarios, and can be used to further verify the adaptability of the proposed method in complex remote sensing scenes.

FBV-Fusion [43] is a dedicated method proposed for remote sensing tiny object detection. It enhances target region representation through mechanisms such as foreground-background view fusion, target mask generation, and skip convolution, and it also constructs the RS-STOD dataset in the corresponding article. Therefore, it is a representative comparison method on this dataset.

DCFL + RPUN-L [44] is an improved method designed for rotated tiny object detection. Its core idea is to alleviate the imbalance in sample matching and supervision assignment for tiny objects through dynamic priors and a coarse-to-fine label assignment mechanism, making it a highly representative approach for remote sensing tiny object detection tasks.

E. Comparative Experimental Analysis

To comprehensively validate the effectiveness and superiority of the proposed method, this article systematically compares it with a variety of representative high-performing detection models under different experimental settings. The corresponding results are presented in Tables II–V. Unless otherwise specified, the symbol "—" in Tables II–V indicates that the metric was not reported in the original article; due to the lack of publicly available code, we were unable to reproduce it reliably and therefore did not supplement it. The symbol "−1" indicates that no valid samples were available for the corresponding scale in COCO evaluation, and thus the metric could not be computed effectively.

It should be noted that the small, medium, and large categories in COCO evaluation are defined by bounding-box area rather than by object category. Let A be the bounding-box area. Then, the three scales correspond to $A < 32^2$, $32^2 \le A < 96^2$, and $A \ge 96^2$, respectively. Since large-object samples are extremely rare or even absent in remote sensing small-object datasets, some large-scale metrics may appear as "−1."

1) Comparative Analysis on the VisDrone2019-DET: To evaluate the detection performance in complex UAV scenarios, VisDrone2019-DET is selected as the primary dataset, and the developed method is compared with the baseline model RT-DETR. The results are shown in Table II. Overall, the proposed method achieves better detection results on VisDrone2019-DET.

The proposed method achieves 33.7% AP, 55.0% AP_{50}, and 34.4% AP_{75} on the VisDrone2019-DET validation set, ranking first among all comparison methods. Compared with the baseline RT-DETR, the proposed method improves AP, AP_{50}, and AP_{75} by 3.2, 4.0, and 3.6 percentage points, respectively, indicating that the proposed improvement strategies can effectively enhance the model's detection performance in complex aerial scenes. Furthermore, the detection performance of the proposed method on small and medium targets is also significantly improved: AP_s increases from 21.2% to 24.9%, and AP_m rises from 42.1% to 45.1%, indicating that the method can more fully exploit shallow detail information and multiscale contextual semantics, and is therefore more adaptable to dense small targets.

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Compared with other advanced methods, the proposed method also exhibits strong competitiveness. Compared with PSWP-DETR, the proposed method achieves higher overall detection accuracy with only 48.9-M parameters, improving AP by 1.2 percentage points, and improving AP_{50}, AP_{75}, and AP_s by 0.9, 1.9, and 2.6 percentage points, respectively, while reducing the parameter count by about 50.5%. In addition, the overall AP of the proposed method is also better than that of CAEM-DETR, CSFPR-RTDETR, and multiple YOLO-series and DETR-series comparison methods, which verifies its effectiveness in UAV small-object detection tasks. It should be pointed out that the proposed method still incurs a certain cost in terms of the large-object metric AP_l and inference speed, where AP_l is 53.2% and FPS is 16.9. Overall, while maintaining a favorable model size, the proposed method significantly improves detection accuracy on the VisDrone2019-DET dataset, especially showing more obvious advantages in small-object detection. Although its inference speed is slightly lower than that of the baseline, its accuracy advantage is noteworthy.

2) Comparative Analysis on the RSOD Dataset: As shown in Table III, the proposed method achieves the best overall detection performance on the RSOD dataset. Specifically, compared with the baseline RT-DETR, the proposed method improves overall AP by 0.7 percentage points, AP_{50} by 0.1 percentage points, and AP_{75} by 0.9 percentage points, indicating that the improved model not only enhances overall detection accuracy, but also exhibits better target localization capability under stricter IoU thresholds. Compared with YOLO-series methods, the proposed method still achieves a 1.0 percentage point AP gain over the best-performing YOLOv11x, indicating stronger overall detection advantages in remote sensing scenes.

From the detection results for targets of different scales, the proposed method performs particularly well on small and large targets. The AP_s and AP_l of the proposed method reach 38.9% and 75.3%, respectively, which are 1.8 and 0.8 percentage points higher than those of the baseline RT-DETR, and 4.3 and 0.9 percentage points higher than the best YOLO-series results, respectively. This indicates that the proposed method can preserve shallow detail information more effectively and strengthen multiscale contextual modeling ability, thereby improving the recognition of dense small targets and targets with significant scale variation. It should be noted that the proposed method reaches 72.7% on AP_m, which is 0.5 percentage points lower than RT-DETR, but the overall AP, AP_{75}, and small-target metrics are all improved, indicating that the method further strengthens small-object detection and high-precision localization while basically maintaining medium-scale target detection capability.

In terms of model complexity, the developed method contains 48.9-M parameters, which is slightly lower than the 49.8 M of RT-DETR, indicating that the introduced modifications do not impose a significant additional model burden. However, its inference speed decreases from 26.9 to 12 FPS, suggesting that the improvement in accuracy is achieved at the cost of a certain reduction in speed. Overall, on the RSOD dataset, the developed method achieves superior detection accuracy with a slightly lower parameter count, and shows notable advantages particularly in small-object detection and localization accuracy under high IoU thresholds, thereby demonstrating its effectiveness in remote sensing object detection tasks.

3) Comparative Analysis on the USOD Dataset: As shown in Table IV, the proposed method achieves 37.7% AP, 92.9% AP_{50}, and 17.8% AP_{75} on the USOD dataset, outperforming the baseline RT-DETR on all major accuracy metrics. Specifically, compared with RT-DETR, the proposed method improves AP, AP_{50}, and AP_{75} by 1.0, 1.0, and 1.3 percentage points, respectively. At the same time, the detection performance on small and medium targets also improves from 36.7% and 45.7% to 37.6% and 47.3%, respectively, indicating that the proposed method is better adapted to both ultrasmall targets and medium-scale targets.

Compared with the other comparison methods, the proposed method also demonstrates strong overall competitiveness. Compared with FFCA-YOLO, the proposed method improves AP and AP_{50} by 2.8 and 2.2 percentage points, respectively, showing better overall detection performance. Compared with multiple YOLO-series methods, the proposed method also maintains an advantage in overall AP, AP_{50}, and the medium-target detection metric, indicating that the proposed detail-semantic collaborative enhancement strategy can more effectively adapt to the characteristics of the USOD dataset, where targets are small in scale and backgrounds are complex. Although the FPS of the proposed method is 6.5, which is lower than that of some lightweight one-stage models, its stable performance in protection tasks.

4) Comparative Analysis on the RS-STOD Dataset: The comparison results on the RS-STOD dataset are presented in Table V. Among all competing methods, the proposed method achieves the best detection performance, with an $AP^{val}_{50:95}$ of 52.6. Compared with the strongest current baseline, RT-DETR, the proposed method improves $AP^{val}_{50:95}$, $AP^{val}_{0.50}$, and $AP^{val}_{0.75}$ by 3.6, 3.0, and 5.2 percentage points, respectively, indicating that it not only improves overall detection accuracy but also significantly enhances object localization capability. Furthermore, the proposed method shows a more pronounced advantage in small-object detection: AP^{val}_S increases from 47.2 to 51.2, whereas AP^{val}_M rises only slightly to 69.6. The more pronounced improvements in AP_S indicate that the proposed method is more beneficial for small-object detection. This is consistent with the design objective of introducing high-resolution detail compensation and multiscale context modeling. Notably, the proposed method contains 48.9-M parameters, which is slightly lower than the 49.8 M of RT-DETR, indicating that the performance improvement does not rely on an increase in model size. However, its inference speed drops to 9.72 FPS, suggesting that the additional feature enhancement and branch computation introduce a certain inference overhead, leaving room for further lightweight optimization in future work.

F. Visual Verification of the Conflict Caused by Shallow Feature Sharing

To visually analyze the representational conflict that may arise from directly sharing shallow features, a typical dense

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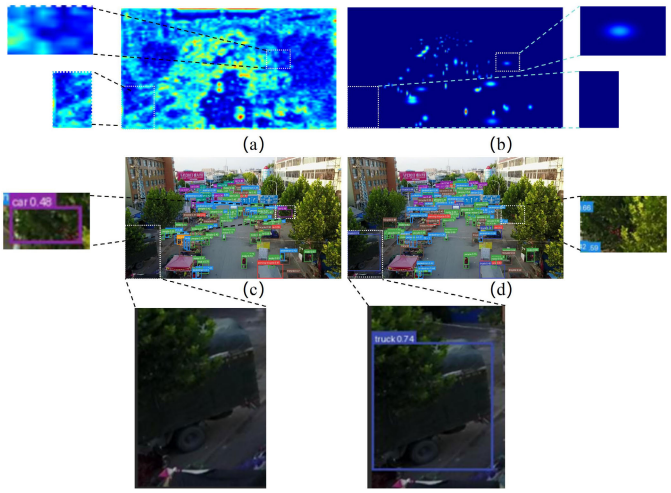


Fig. 5. Visual verification of the conflict caused by shallow feature sharing.

small-object scene from the VisDrone2019-DET dataset is selected for visualization, as shown in Fig. 5. Specifically, Fig. 5(a) shows the response heatmap of RT-DETR at the shallow shared feature layer C_3, and Fig. 5(b) shows the response heatmap of the RT-DETR decoder output. The heatmaps are obtained by averaging the corresponding feature maps along the channel dimension and then normalizing them, so as to reflect the response distribution of each layer over target and background regions. Fig. 5(c) and (d) presents the detection results of RT-DETR and the proposed DSH-DETR, respectively. For better observation, typical local regions are enlarged, and the false positives and missed detections are marked in the figure.

As can be seen from Fig. 5(a), the shallow shared features of RT-DETR exhibit relatively scattered response distributions in complex scenes. In addition to the target regions, some background texture regions are also strongly activated, while the responses to certain tiny or occluded targets remain relatively weak. This indicates that when shallow features are required to simultaneously preserve local details and support subsequent semantic transmission, their representations are more easily disturbed by background clutter, making it difficult to sufficiently highlight the edge and structural information associated with small objects.

Further observation of Fig. 5(b) shows that although the responses to some salient targets are enhanced after the subsequent encoding and decoding processes, unnecessary activations are still retained in several background regions, and the responses to weak small objects remain insufficient. This suggests that inadequate representations at the shallow stage can be further propagated to the later prediction stage, thereby affecting the model's ability to discriminate and localize small objects in complex scenes.

The effect can be understood more intuitively by comparing the detection results in Fig. 5(c) and (d). RT-DETR is more likely to produce spurious responses in complex background regions, leading to false detections. At the same time, it fails to detect some weak or occluded small objects completely, resulting in missed detections or inaccurate localization. In

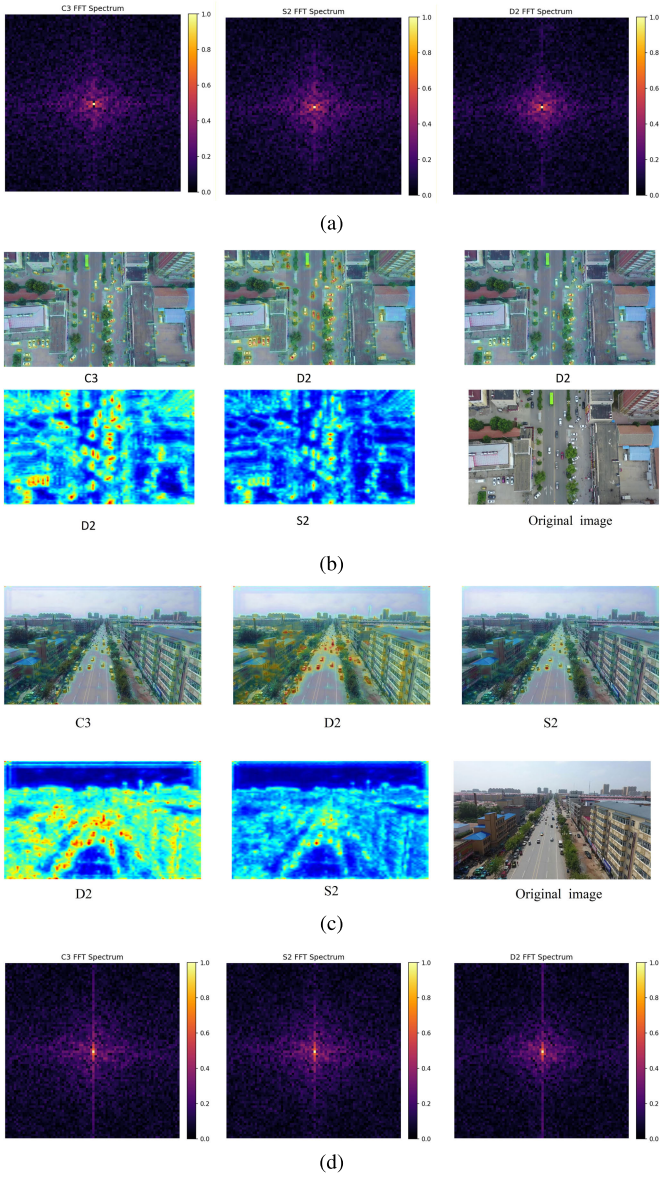


Fig. 6. Frequency-domain and spatial-response visualization of D_2 and S_2 after decoupling from C_3.

contrast, the proposed DSH-DETR can suppress background interference more effectively while preserving target structural information more completely, thus achieving better small-object detection performance and target separation ability in the same scene.

These observations indicate that directly sharing shallow features can, to some extent, introduce representational competition between detail preservation and semantic modeling, thereby degrading detection performance in complex remote sensing scenes. By introducing a detail–semantic decoupling mechanism at the encoding stage, the proposed method enables shallow features to be optimized separately along the detail path and the semantic path, thereby alleviating the conflict in shared representations and providing more stable and effective feature support for subsequent small-object detection.

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G. Rationality Analysis of D2/S2 Decoupling

To further investigate the effect of decoupling the shallow feature C_3 into the detail branch D_2 and the semantic branch S_2, Fig. 6 presents the frequency-domain and spatial-response visualizations in two representative scenes. Specifically, Fig. 6(a) shows the FFT spectra of C_3, S_2, and D_2 for the first scene, and Fig. 6(b) shows the corresponding spatial-response heatmaps. Fig. 6(c) presents the spatial-response heatmaps for the second scene, while Fig. 6(d) shows the corresponding FFT spectra of C_3, S_2, and D_2. As shown in Fig. 6(a), in the first top-view scene, compared with the original C_3 feature, S_2 exhibits a more concentrated energy distribution around the low-frequency center, indicating that the semantic branch tends to preserve global structure and contextual information while suppressing redundant local texture fluctuations. In contrast, D_2 retains relatively stronger responses in the mid- and high-frequency regions, suggesting greater sensitivity to local details such as vehicle contours, road boundaries, building edges, and texture transitions. The spatial-response maps in Fig. 6(b) further support this observation: D_2 produces sharper and more localized activations around densely distributed vehicles, road boundaries, and other fine-grained local regions, which is beneficial for preserving small-object details and enhancing localization-related cues, whereas S_2 yields smoother and more spatially coherent responses, with greater emphasis on road distribution, building layout, and surrounding scene context.

A similar trend can also be observed in the second representative scene. As shown in Fig. 6(d), in the street scene with stronger perspective effects and a more dominant road direction, S_2 still exhibits more compact low-frequency energy and a more stable distribution along the principal scene structure, indicating that it mainly captures global semantic information such as road continuity, building arrangement, and overall scene coherence. In contrast, D_2 preserves more pronounced mid- and high-frequency components and remains more responsive to vehicle edges, roadside structures, building contours, and local texture variations. The spatial-response maps in Fig. 6(c) further show that D_2 focuses more on vehicle-populated regions, road boundaries, and fine-grained structures along the street corridor, whereas S_2 highlights road orientation, spatial depth, and overall structural semantics. These consistent observations across the two representative scenes demonstrate that the proposed decoupling does not simply duplicate the original shallow feature, but instead drives it toward two complementary roles, namely, detail preservation and semantic modeling, thereby alleviating the representational conflict in shared shallow features and providing a clearer and more discriminative feature basis for subsequent small-object detection.

H. Ablation Studies

1) Ablation Study of the Main Components: To verify the effectiveness of each improved module, ablation experiments are conducted on the VisDrone2019-DET dataset. The results are shown in Table VI. The baseline model achieves 30.5% AP, 51.0% AP_{50}, and 30.8% AP_{75} on the validation set. After introducing the HRSM module, the model performance improves significantly: AP increases to 33.0%, which is 2.5 percentage points higher than the baseline, while AP_{50}, AP_{75}, and AP_s improve by 3.5, 2.7, and 2.9 percentage points, respectively, indicating that the high-resolution semantic enhancement strategy can effectively supplement shallow detail information and improve the model's capability for small-object detection.

When only the MPSEM module is introduced, the model's AP increases to 30.9%, which is 0.4 percentage points higher than the baseline, while AP_l rises from 55.5% to 59.8%, indicating that this module has a certain promoting effect on high-level semantic modeling and large-object representation, but its improvement to overall detection accuracy is relatively limited. Furthermore, when HRSM and MPSEM are introduced simultaneously, the model achieves the best performance: AP, AP_{50}, and AP_{75} reach 33.7%, 55.0%, and 34.4%, respectively, which are 3.2, 4.0, and 3.6 percentage points higher than the baseline; meanwhile, AP_s and AP_m reach 24.9% and 45.1%, respectively, which are 3.7 and 3.0 percentage points higher than the baseline. These results indicate that the two modules complement each other well in functionality: the former focuses on enhancing high-resolution details and shallow semantic interaction, while the latter further strengthens multiscale context and positional-information modeling. Their synergy significantly improves the model's detection performance in complex UAV scenes.

In addition, from the perspective of model complexity, after introducing HRSM, the parameter count decreases to 47.3 M, and after adding MPSEM simultaneously, the parameter count becomes 48.9 M, which is basically comparable to the 49.8 M of the baseline, indicating that the proposed method achieves better detection performance without significantly increasing model size. Although inference speed decreases from 32.4 to 16.9 FPS, this cost is acceptable considering the clear improvement in detection accuracy, especially for small-object detection.

2) Scale-Wise Recall Analysis: To avoid simply attributing the improvement in overall AP to gains in small-object performance, we further report scale-wise recall results for different

TABLE VI
ABLATION STUDY ANALYSIS ON VISDRONE2019-DET

baseline	HRSM	MPSEM	AP^{val}	$AP^{val}_{0.50}$	$AP^{val}_{0.75}$	AP^{val}_s	AP^{val}_m	AP^{val}_l	FPS	Params (M)
✓			30.5	51.0	30.8	21.2	42.1	55.5	32.4	49.8
✓	✓		33.0	54.5	33.5	24.1	43.3	55.8	11.0	47.3
✓		✓	30.9	51.5	30.9	21.6	42.6	59.8	21.8	52.0
✓	✓	✓	33.7	55.0	34.4	24.9	45.1	53.2	16.9	48.9

TABLE VII
ABLATION STUDY OF SCALE-WISE RECALL UNDER DIFFERENT MODULE SETTINGS

Baseline	HRSM	MPSEM	AR^{val}	AR^{val}_S	AR^{val}_M	AR^{val}_L
✓			45.6	36.5	58.5	71.7
✓	✓		47.9	39.6	59.8	69.5
✓		✓	45.4	36.2	58.1	73.4
✓	✓	✓	49.1	41.2	60.5	72.5

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TABLE VIII

ABLATION STUDY OF STRIP-CONVOLUTION DIRECTIONAL CONFIGURATIONS IN MPSEM

baseline	HRSM	H-Strip	V-Strip	AP^{val}	$AP^{val}_{0.50}$	$AP^{val}_{0.75}$	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}	FPS	Params (M)
✓	✓	✓		33.1	54.7	33.7	24.4	44.5	54.4	12.6	48.4
✓	✓		✓	32.9	54.2	32.9	23.8	43.8	57.2	11.1	48.4
✓	✓	✓	✓	33.1	54.7	33.9	24.1	44.6	58.5	12.7	48.4

module settings, as summarized in Table VII. Compared with the baseline, introducing HRSM alone improves AR^{val} from 45.6 to 47.9, where AR^{val}_{S} increases significantly from 36.5 to 39.6, corresponding to a gain of 3.1 points, while AR^{val}_{M} also rises from 58.5 to 59.8. These results indicate that high-resolution feature compensation can effectively alleviate missed detections of small objects and is an important factor in improving small-object detection performance.

By contrast, when only MPSEM is introduced, the change in overall recall is relatively limited: AR^{val}_{S} slightly decreases from 36.5 to 36.2, whereas AR^{val}_{L} increases from 71.7 to 73.4, suggesting that MPSEM is more inclined to enhance large-object representation and global contextual modeling. Furthermore, when HRSM and MPSEM are jointly used, the model achieves the best overall recall performance, with AR^{val}, AR^{val}_{S}, and AR^{val}_{M} reaching 49.1, 41.2, and 60.5, respectively. This indicates that HRSM mainly improves small-object recall, whereas MPSEM provides complementary contextual and multiscale modeling capability, and the two modules exhibit good complementarity.

Based on the above results, the previous statement that the overall AP improvement mainly comes from high-resolution feature compensation is revised to a more cautious conclusion: high-resolution feature compensation is one of the major sources of performance improvement, and its benefit is mainly reflected in the improvement of small-object recall. Meanwhile, MPSEM plays a supporting role in enhancing feature representation across different scales, and the final overall performance improvement is achieved through their joint effect. Furthermore, existing multiscale enhancement methods that still operate on shared shallow features often mainly improve contextual compensation or larger scale target representation, while their ability to alleviate missed detections of small objects and the competition within shallow detail representations remains relatively limited. In contrast, the proposed method first changes the way shallow features are utilized through detail-semantic dual-branch routing, and then performs targeted optimization for detail preservation and weakened low-level semantics, thereby enabling more stable and task-oriented performance gains in complex remote sensing scenes.

3) Directional Analysis of Strip Convolution: To further analyze the directional modeling capability of strip convolution in MPSEM, three settings are considered, namely, using only horizontal strip convolution (H-Strip), using only vertical strip convolution (V-Strip), and using both simultaneously. The results are presented in Table VIII. It can be seen that using only H-Strip yields an AP of 33.1, which is slightly higher than the 32.9 obtained by using only V-Strip. In addition, H-Strip also performs better in $AP^{val}_{0.50}$, $AP^{val}_{0.75}$, AP^{val}_{S}, and AP^{val}_{M}, indicating that under the current data distribution, horizontal contextual aggregation is more effective for detecting small and medium-scale objects.

TABLE IX

ABLATION STUDY OF DCAU PLACEMENT AT DIFFERENT FEATURE LEVELS

Location	DCAU	AP^{val}	$AP^{val}_{0.50}$	$AP^{val}_{0.75}$	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}	FPS	Params (M)
C3	✓	31.0	52.0	31.1	21.9	42.7	55.5	14.9	43.8
C4	✓	30.8	51.5	30.6	21.5	42.4	54.4	20.2	47.0
C5	✓	30.8	51.4	30.9	21.6	42.2	57.7	19.9	59.9

TABLE X

ACCURACY-EFFICIENCY TRADEOFF ANALYSIS

baseline	HRSM	Strip	Dilated	DCAU	AP^{val}	$AP^{val}_{0.50}$	$AP^{val}_{0.75}$	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}	FPS	Params (M)
✓	✓	✓			33.1	54.7	33.9	24.1	44.6	58.5	12.7	48.4
✓	✓		✓		32.9	54.5	33.5	23.9	44.4	55.4	9.1	49.2
✓	✓			✓	33.1	54.8	33.7	24.3	44.3	56.1	13.9	48.4
✓	✓	✓	✓	✓	33.7	55.0	34.4	24.9	45.1	53.2	16.9	48.9

On the other hand, when only V-Strip is used, the model achieves a higher AP^{val}_{L} of 57.2, which is better than the 54.4 obtained by H-Strip, suggesting that vertical long-range dependency modeling provides a certain advantage for large-scale targets. Furthermore, when both H-Strip and V-Strip are introduced simultaneously, the model achieves the best results, with $AP^{val}_{0.75}$, AP^{val}_{M}, and AP^{val}_{L} reaching 33.9, 44.6, and 58.5, respectively, while the number of parameters remains nearly unchanged. These results indicate that strip convolution is sensitive to directional information, but more importantly, there exists clear complementarity between horizontal and vertical directions. The dual-direction design can more fully model the anisotropic structures of remote sensing targets, thereby improving localization precision and scale adaptability.

4) Effect of DCAU Placement: To verify the effect of DCAU at different feature levels, it is embedded into the C_3, C_4, and C_5 layers for comparison, and the results are shown in Table IX. It can be observed that when DCAU is placed at C_3, the model achieves the highest overall detection accuracy, where AP^{val}, $AP^{val}_{0.50}$, and $AP^{val}_{0.75}$ reach 31.0, 52.0, and 31.1, respectively, all outperforming the results obtained when it is placed at C_4 or C_5. This indicates that CA can more effectively enhance spatial positional information and channel responses on shallow high-resolution features, which is particularly beneficial for distinguishing dense small objects and medium-scale targets in remote sensing scenes.

Although placing DCAU at C_5 yields a higher AP^{val}_{L} of 57.7, the overall AP is slightly lower, and the parameter increase is more noticeable. Considering both detection accuracy and model complexity, DCAU is finally deployed at the C_3 layer in this work.

5) Accuracy-Efficiency Tradeoff Analysis: To further analyze the accuracy-efficiency tradeoff of each submodule, strip convolution, dilated convolution, and DCAU are introduced separately into the same baseline framework and quantitatively compared. The results are shown in Table X. It can be seen that when only strip convolution is introduced, the model achieves AP^{val} = 33.1 with 48.4-M parameters and 12.7 FPS, indicating that this module can provide stable detection gains with relatively small additional overhead.

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TABLE XI
REDUNDANCY ANALYSIS

baseline	HRSM	upsample	AP^{val}	$AP^{val}_{0.50}$	$AP^{val}_{0.75}$	AP^{val}_{S}	AP^{val}_{M}	AP^{val}_{L}	FPS	Params (M)
Baseline	✓	✓	33.0	54.5	33.5	24.1	43.3	55.8	11.0	47.3
Baseline	✓	×	33.2	54.0	34.0	24.2	44.4	50.9	15.7	47.8

By contrast, when only dilated convolution is introduced, the model achieves AP^{val} = 32.9, while the number of parameters increases to 49.2 M and FPS decreases to 9.1, suggesting that dilated convolution imposes a more noticeable computational and memory burden and offers relatively limited efficiency gains when used alone. When only DCAU is introduced, the model achieves AP^{val} = 33.1, $AP^{val}_{0.50}$ = 54.8, and AP^{val}_{S} = 24.3, while maintaining 13.9 FPS, indicating that this module can effectively enhance spatial localization under relatively low complexity and is more favorable for small-object detection. Furthermore, when strip convolution, dilated convolution, and DCAU are jointly employed, the model achieves the best overall performance, where AP^{val}, $AP^{val}_{0.75}$, AP^{val}_{S}, and AP^{val}_{M} reach 33.7, 34.4, 24.9, and 45.1, respectively, demonstrating that the three components exhibit good complementarity in feature modeling.

6) Redundancy Analysis: To analyze whether the decrease in inference speed is related to redundant computation, the necessity of the upsampling operation in HRSM is further investigated, and the results are shown in Table XI. When the upsampling operation is retained in HRSM, the model achieves AP^{val} = 33.0, $AP^{val}_{0.50}$ = 54.5, and AP^{val}_{L} = 55.8. Although removing the upsampling operation leads to slight improvements in AP^{val}, $AP^{val}_{0.75}$, AP^{val}_{S}, AP^{val}_{M}, and FPS, the version with upsampling performs better in $AP^{val}_{0.50}$ and especially AP^{val}_{L}, indicating that upsampling helps feed high-level semantic information back into the high-resolution feature space more sufficiently, thereby enhancing the representation capability for large objects and improving overall detection performance under low-threshold conditions.

Meanwhile, the version with upsampling contains 47.3-M parameters, which is slightly lower than the 47.8-M parameters of the version without upsampling. Therefore, from the design objective of HRSM, the upsampling operation should not be simply regarded as ineffective redundancy. Instead, it is not pure redundancy; however, from the perspective of the overall accuracy–efficiency tradeoff, its cost-effectiveness is relatively limited. Although this design incurs a certain loss in inference speed, it provides more complete semantic feedback and more stable large-object detection performance, and is therefore retained in the final model.

It is worth emphasizing that while Params mainly indicate storage footprint, FPS is affected by a combination of feature map resolution, branch count, operator types, memory access behavior, and cross-scale fusion. Hence, a modest drop in parameters does not guarantee faster inference. In our work, the observed speed reduction stems mainly from extra operations in high-resolution feature spaces and multibranch interactions, not from a significant growth in model weights.

I. Visualization Comparison Analysis

1) To further evaluate the detection performance and generalization ability of DSH-DETR, visual comparison results on the VisDrone2019-DET, RSOD, USOD, and RS-STOD datasets are presented and discussed. Fig. 7 presents the quantitative comparison results on the VisDrone2019-DET dataset. As shown in Fig. 7, the proposed method outperforms the baseline RTDETR in terms of AP, AP50, AP75, APs, and APm. As shown in Fig. 8, typical complex scenes from the VisDrone2019-DET dataset are selected to visually compare and analyze the detection results of different methods. In this scene, targets are densely distributed, scale differences are obvious, and there are a large number of small targets in the distant intersection area. At the same time, there are interference factors such as occlusion and complex background textures on both sides of the road, which place high demands on the detector's fine-grained representation ability and multiscale modeling capability.

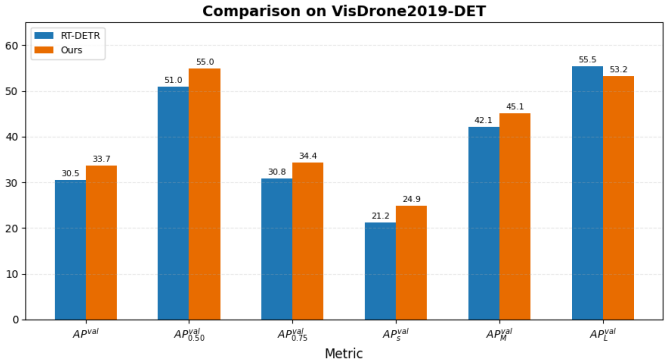


Fig. 7. Comparative analysis on VisDrone2019-DET.

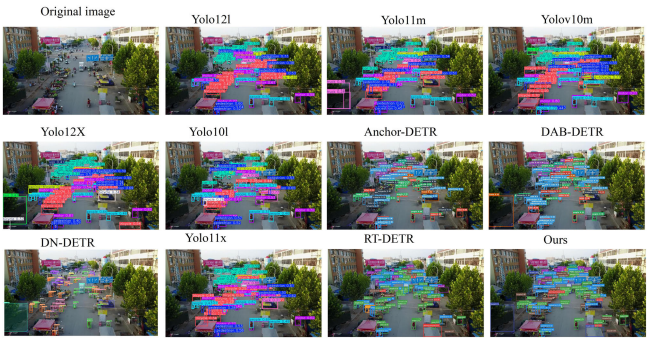


Fig. 8. Visualization comparison results on VisDrone2019-DET.

From the detection results of the YOLO-series methods, it can be seen that YOLO11x, YOLO12l, YOLO12x, YOLOv10m, YOLO11m, and YOLO10l can basically detect relatively salient foreground targets in the image, but they still show obvious missed detections, duplicate detections, and category confusion in the distant intersection area and densely populated regions on both sides of the road. In particular, in areas where neighboring targets such as small pedestrians, bicycles, and tricycles are densely clustered, different methods generally suffer from insufficient target separation ability, and some predicted boxes overlap severely or are inaccurately localized.

For DETR-series methods such as Anchor-DETR, DAB-DETR, and DN-DETR, although their detection results are relatively stable on some salient targets, they still show insufficient recall in distant small-target regions, and some

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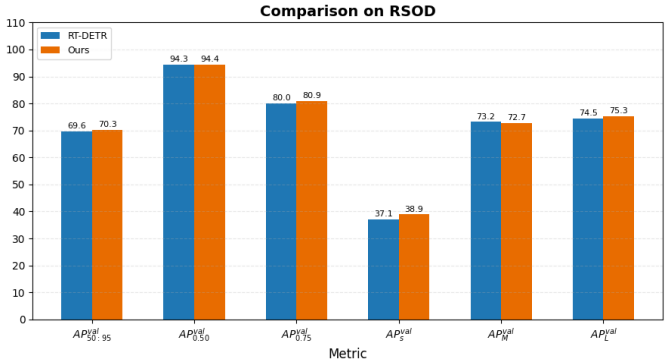


Fig. 9. Comparative analysis on the RSOD dataset.

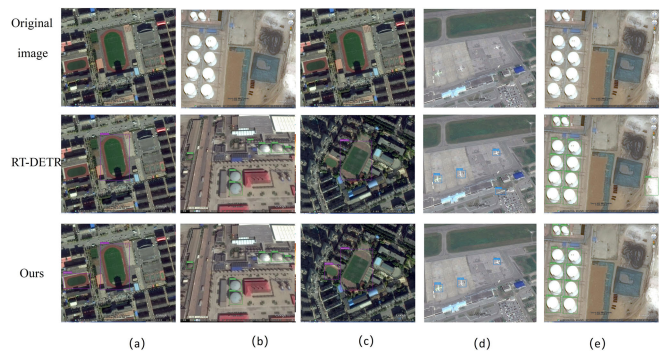


Fig. 10. Visualization comparison results on RSOD.

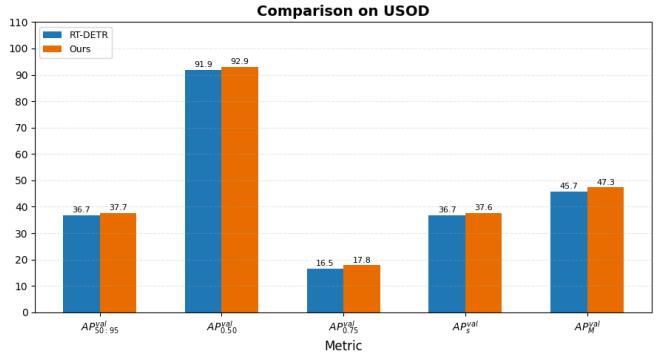


Fig. 11. Comparative analysis on the USOD dataset.

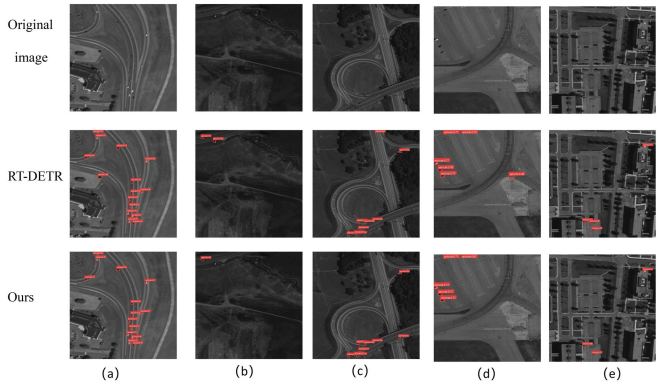


Fig. 12. Visualization comparison results on USOD.

marginal targets and weak-texture targets fail to be effectively recognized. At the same time, under complex background interference, some target boxes are oversized, or their category predictions are not accurate enough, indicating that such methods still have certain limitations in preserving high-resolution details and enhancing the semantics of small targets.

These visual results are consistent with the quantitative results in Fig. 7. These results show that the proposed method can make more effective use of shallow detail information and high-level semantic information, thereby achieving better detection performance in scenes with dense small targets, long-distance targets, and complex backgrounds. Overall, the proposed method shows more obvious advantages in reducing missed detections, suppressing false detections, and improving the separation ability of dense targets.

2) To further verify the detection performance of the proposed method on the RSOD dataset, Fig. 9 presents the comparison results between RT-DETR and the proposed method on various metrics, and Fig. 10 shows the visual detection results of the two methods in six typical scenes. Each column corresponds to one scene, while the three rows show the original image, the detection results of RT-DETR, and the detection results of DSH-DETR, respectively. Specifically, (a) presents a playground scene, (b) presents a storage-tank scene, (c) presents another playground scene, (d) presents an airport scene with aircraft, and (e) presents another storage-tank scene. As can be seen from Fig. 9, the proposed method improves $AP_{50:95}$, AP_{50}, AP_{75}, AP_s, and AP_l from 69.6, 94.3, 80.0, 37.1, and 74.5 to 70.3, 94.4, 80.9, 38.9, and 75.3, respectively, corresponding to gains of 0.7, 0.1, 0.9, 1.8, and 0.8 percentage points. Only the AP_m metric decreases slightly from 73.2 to 72.7. Overall, the proposed method still exhibits better overall detection capability on the RSOD dataset, especially with more obvious improvements in small-object detection and localization accuracy under high IoU thresholds.

In Fig. 10(a), the scene contains both a large-scale playground and a relatively small playground target. RT-DETR detects only the main large-scale target and fails to effectively identify the smaller playground on the left. In contrast, the proposed method detects both playground regions simultaneously, indicating that the improved model has better target integrity and small-scale target perception capability in multitarget scenes.

In Fig. 10(b), multiple aircraft targets are present in the airport scene, and the runway and apron textures in the background are relatively complex. Although RT-DETR can detect the main aircraft targets, it produces extra responses in the lower right region. The proposed method, while preserving the detection results of the three real aircraft targets, suppresses these redundant predictions, indicating better false-positive control capability under complex backgrounds.

From the visualization results in Fig. 10(a)–(e), it can be seen that the advantages of the proposed method on the RSOD dataset are mainly reflected in two aspects: first, it can detect multiscale targets in the scene more completely, especially small playgrounds and other small-scale targets that are easily overlooked; second, it can effectively suppress false detections caused by complex backgrounds, highlighted regions, and structural textures. This phenomenon is consistent with the quantitative results in Fig. 9. In particular, the improvements

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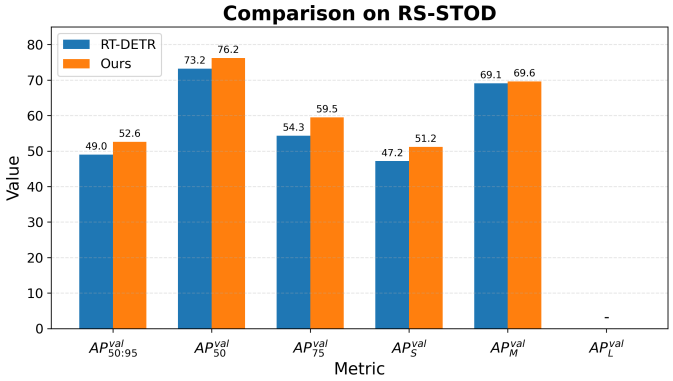


Fig. 13. Quantitative comparison analysis on the RS-STOD dataset.

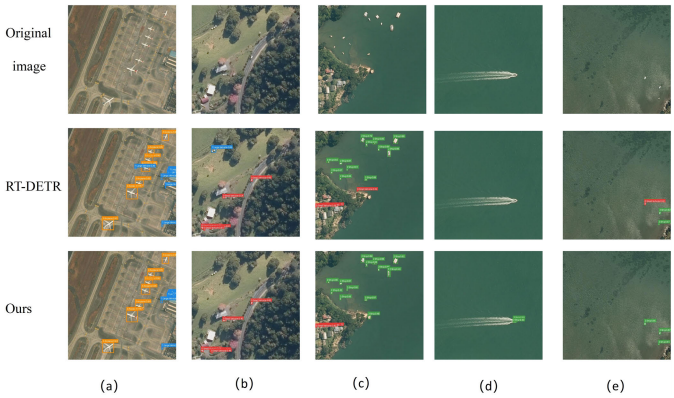


Fig. 14. Visual comparison results on the RS-STOD dataset.

in AP_s and AP_{75} further indicate that the improved model not only enhances small-target detection ability, but also improves target localization accuracy. Although there is a slight decrease in the AP_m metric, the overall results show that the proposed method has better overall detection performance and stronger scene adaptability on the RSOD dataset.

3) As further verification of the effectiveness of the proposed method on the USOD dataset, Fig. 11 presents the quantitative comparison results between RT-DETR and the proposed method, while Fig. 12 shows the visual detection results of the two methods in typical scenes. Each column represents one representative scene, and the three rows correspond to the original image, the RT-DETR results, and the DSH-DETR results, respectively. Specifically, (a)-(e) show a vehicle-dense road scene, a low-contrast scene, a highway-interchange scene, a parking-lot scene, and a built-up scene with parked vehicles, respectively. As shown in Fig. 11, the proposed method outperforms RT-DETR on all evaluation metrics. Specifically, $AP^{val}_{50:95}$, AP^{val}_{50}, AP^{val}_{75}, AP^{val}_{S}, and AP^{val}_{M} are improved by 1.0, 1.0, 1.3, 0.9, and 1.6 percentage points, respectively. These results indicate that the proposed method exhibits stable advantages in overall detection performance, localization accuracy under high IoU thresholds, and the detection of small- and medium-scale targets.

In Fig. 12(a), vehicle targets are densely distributed, and the distances between adjacent targets are small, causing RT-DETR to produce many overlapping responses in local regions. In contrast, the proposed method effectively suppresses duplicate predictions while retaining the main targets, yielding more compact detection results and demonstrating better discrimination ability in dense scenes. In Fig. 12(b), the overall texture of the region is weak, and the contrast is low, making the vehicle-target features less salient. RT-DETR produces multiple weak responses in the upper left area, whereas the proposed method retains only more reliable target predictions and reduces invalid detections, demonstrating stronger background suppression and anti-interference capability.

4) To further verify the effectiveness of the proposed method on the RS-STOD dataset, Fig. 13 presents the quantitative comparison results of RT-DETR and the proposed method on various evaluation metrics, while Fig. 14 shows the visual detection results of the two methods in typical scenes. As shown in Fig. 13, the proposed method improves $AP_{50:95}$, AP_{50}, AP_{75}, AP_S, and AP_M from 49.0, 73.2, 54.3, 47.2, and 69.1 to 52.6, 76.2, 59.5, 51.2, and 69.6, respectively, corresponding to gains of 3.6, 3.0, 5.2, 4.0, and 0.5 percentage points. Among them, the improvements in AP_{75} and AP_S are more evident, indicating that the proposed method not only enhances small-object detection performance in complex remote sensing scenes, but also achieves higher localization accuracy under stricter IoU thresholds. Since the RS-STOD dataset is dominated by small- and medium-scale targets, AP_L is not reported separately.

The visual results in Fig. 14 further demonstrate that the proposed method exhibits better detection stability across different types of complex scenes. Specifically, each column represents a representative scene, while the three rows correspond to the original image, the RT-DETR results, and the DSH-DETR results, respectively. From left to right, (a)-(e) show an airport scene with aircraft, a road scene with vegetation interference, a near-shore scene with densely distributed ships, an open-water scene with isolated boats, and a low-contrast sea-clutter scene with tiny ships, respectively. In airport scenes, where aircraft and vehicles coexist with large-scale variation and complicated paved-texture backgrounds, the proposed method provides more complete target responses. In road scenes with vegetation interference, the proposed method preserves the main vehicle targets while effectively suppressing redundant predictions. In near-shore dense ship scenes, the proposed method shows better separation capability for adjacent small objects, leading to more complete detection results. In open-water scenes with weak texture, the baseline model shows weak responses to isolated targets, whereas the proposed method still achieves stable detection. Under low-contrast sea-clutter backgrounds, the proposed method can identify more tiny ships while reducing false detections and category confusion.

By combining the quantitative and qualitative results, it can be concluded that the advantages of the proposed method on the RS-STOD dataset are mainly reflected in two aspects. First, it can exploit shallow detail information more effectively, thereby improving the completeness of detection for dense small objects and weak small objects. Second, it can model multiscale contextual and positional information more effectively, thereby enhancing robustness against complex

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background interference and improving high-precision localization capability. This phenomenon is consistent with the improvement trends observed in $AP_{50:95}$, AP_{75}, and AP_S, which further verifies the effectiveness of the proposed method for remote sensing small-object detection.

V. Complexity Analysis and Lightweight Exploration

To analyze the deployment cost of the proposed method, the number of parameters and inference speed FPS of the baseline model and different module combinations are uniformly measured and compared under the same hardware environment and input-resolution setting. The results show that the baseline RT-DETR has 49.8-M parameters and runs at 32.4 FPS. After introducing HRSM, the number of parameters decreases to 47.3 M, whereas the FPS drops significantly to 11.0. When only MPSEM is introduced, the number of parameters increases to 52.0 M, with an FPS of 21.8. When both HRSM and MPSEM are introduced simultaneously, the model contains 48.9-M parameters and achieves 16.9 FPS. These results indicate that the proposed method remains comparable to the baseline in terms of parameter scale, but its inference speed decreases noticeably, suggesting that the number of parameters alone cannot fully reflect the actual deployment cost of the model.

Further analysis shows that the reduction in inference efficiency is not mainly caused by a significant increase in model parameters, but rather by the additional feature fusion and upsampling operations introduced by the high-resolution semantic reconstruction branch. In addition, the multibranch semantic enhancement module increases convolutional computation on low-level high-resolution feature maps, branch interaction, and positional-attention calibration, thereby introducing higher feature-level computational overhead and memory-access burden.

To further explore the possibility of lightweight optimization, an ablation study is conducted on the necessity of the upsampling operation in HRSM. The results show that after removing this operation, the model FPS increases from 11.0 to 15.7, while AP changes from 33.0 to 33.2 and AP_{75} increases from 33.5 to 34.0. However, AP_{50} and AP_L decrease slightly. These results indicate that the upsampling operation still contributes positively to certain detection metrics, but from the perspective of the overall accuracy-efficiency tradeoff, its cost-effectiveness is relatively limited, making it one of the important factors causing the current decline in inference speed.

Overall, the proposed method achieves better detection accuracy at the cost of a certain reduction in inference efficiency, and is therefore more suitable for accuracy-oriented remote sensing small-object detection scenarios. Further lightweight optimization for real-time deployment remains an important direction for future research.

VI. Conclusion

To address the insufficient utilization of shallow details, the limited semantic representation capability of low-level features, and the tendency of small objects to be missed in remote sensing object detection, an improved RT-DETR method based on detail-semantic decoupling and multiscale coordinate-guided semantic enhancement is developed. Through a detail-semantic decoupled dual-branch routing mechanism, the shallow C_3 features are divided into a detail path and a semantic path. A multiscale coordinate-guided semantic enhancement module is introduced into the semantic path to improve low-level semantic representation through multibranch semantic modeling, adaptive semantic interactive fusion, and coordinate-guided semantic calibration. Furthermore, by incorporating a high-resolution semantic compensation unit, a four-scale detection framework is constructed, enabling effective collaboration between shallow details and deep semantics.

Although the proposed method achieves consistent improvements in detection accuracy, the additional feature-level computational overhead introduced by high-resolution reconstruction and the multibranch enhancement architecture makes the current method more suitable for accuracy-oriented scenarios. For applications where real-time performance is a higher priority, further lightweight optimization is still required. In future work, we will focus on improving the accuracy-efficiency tradeoff by simplifying the high-resolution compensation branch, replacing redundant upsampling operations, designing lightweight attention mechanisms, and pruning unnecessary branches.

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